

ASTE-331a: Spacecraft Systems Engineering

Mechanical Subsystem

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Nov 14th, 2025

Mechanical Subsystem Overview

Mechanical

Introduction

• Overview

• Examples

Constraints & Reqs.

Design

• Structures

• Materials

• Mechanisms

• Configuration

• Loads and

Environments

• Mass Estimation

• Trade Studies

Analysis and Test

Design Process

Case Studies

• Function

- Provides structure to support spacecraft and payload
- Enables deployment mechanisms and separation systems
- Protects against launch and on-orbit stresses
- Protects against radiation, micrometeoroids
- Primary drivers are:
 - Spacecraft type (orbiter vs. lander/rover)
 - Payload (camera vs. sampling system)
 - Overall dry and wet mass
 - Articulated arrays, antennas, instruments
- Also called Structure & Mechanisms, and often includes the harness (i.e., wiring)

• Common Components

- Primary structure
 - Core structure, panels, etc.
- Secondary structure
 - Tank support, fittings, etc.
 - Payload support, booms
- Mechanisms
 - Gimbals, deployables, etc.
- Payload elements (wheels, sample return canisters, etc.); can also go under Payload
- Launch vehicle and spacecraft adapters

• GRAIL Example

- 4 side panels
- Propulsion & payload panels
- Tank support, fittings...
- Solar array deployment mech.
- LV separation system
- Total = ~35 kg



• Key Trades & Analyses

- Configuration, CAD, & FOV design
- Loads and stiffness analyses
- Master equipment list (MEL)
- Mechanism, torque, etc. design and analyses

• Key Parameters

- Mass, power, & cost

• Relationship with other subsystems

- Houses support and environmental protection for all subsystems
- Provides load distribution for ACS and propulsion subsystems
- Provides grounding and harness support for power subsystem
- Provides heat transfer and dissipation for thermal subsystem

Example: Asteroid Sample Return Mission

Mechanical

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• Examples

Constraints & Reqs.

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Analysis and Test

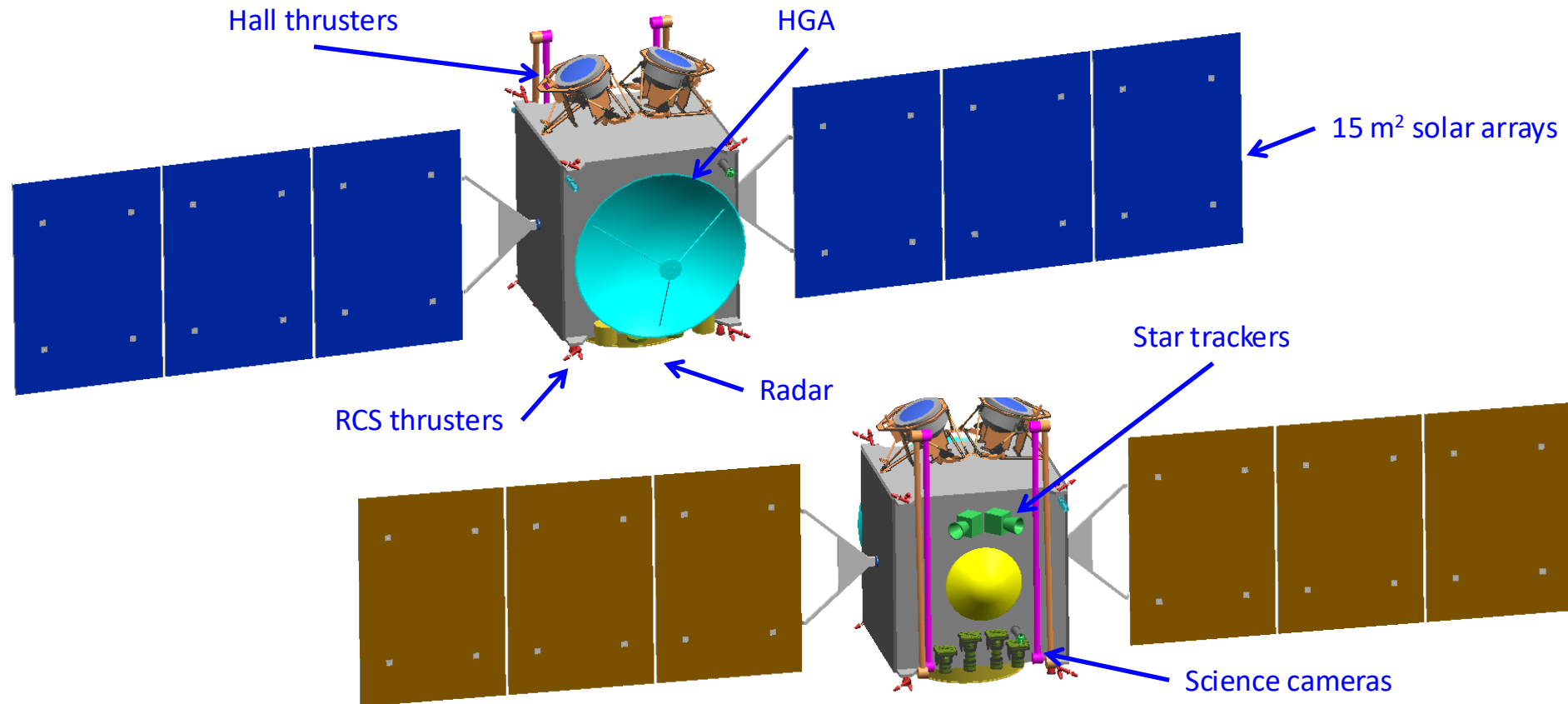
Design Process

Case Studies

• Similar to OSIRIS-Rex

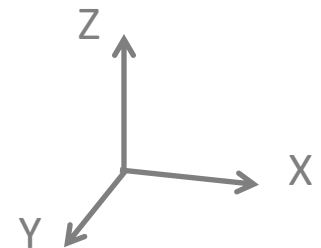
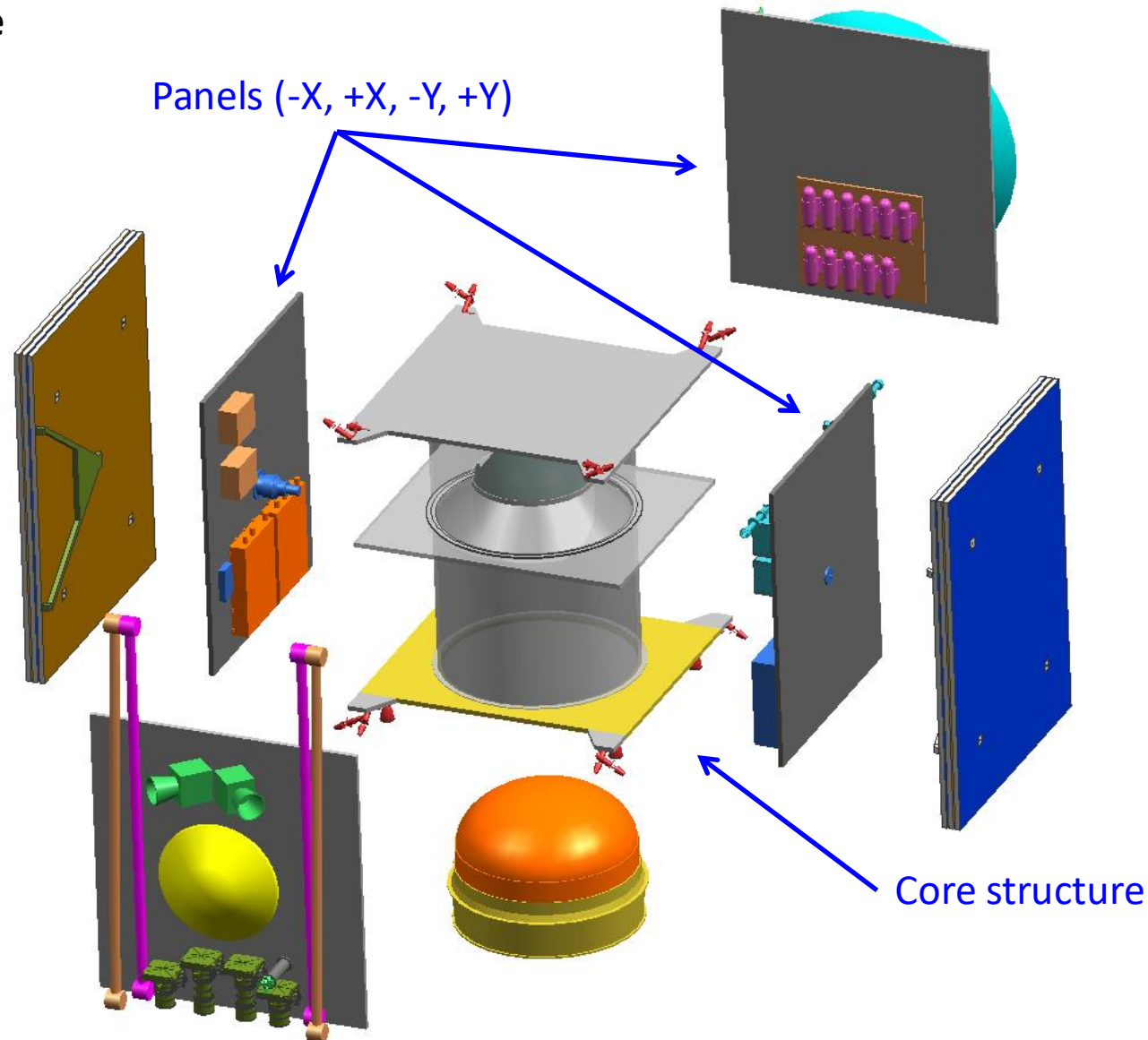
- Deep space mission to a near-Earth asteroid (NEA)
- Collects sample via Touch-And-Go maneuver and stow in Sample Return Canister (SRC)
- Uses electric propulsion (Hall thrusters), including high-voltage power (110 V)

OSIRIS-Rex



Example: Asteroid Sample Return Mission

- Primary Structure



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Introduction

- Overview

- Examples

Constraints & Reqs.

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- Structures

- Materials

- Mechanisms

- Configuration

- Loads and

- Environments

- Mass Estimation

- Trade Studies

Analysis and Test

Design Process

Case Studies

Example: Asteroid Sample Return Mission

Mechanical

Introduction

• Overview

• Examples

Constraints & Reqs.

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• Structures

• Materials

• Mechanisms

• Configuration

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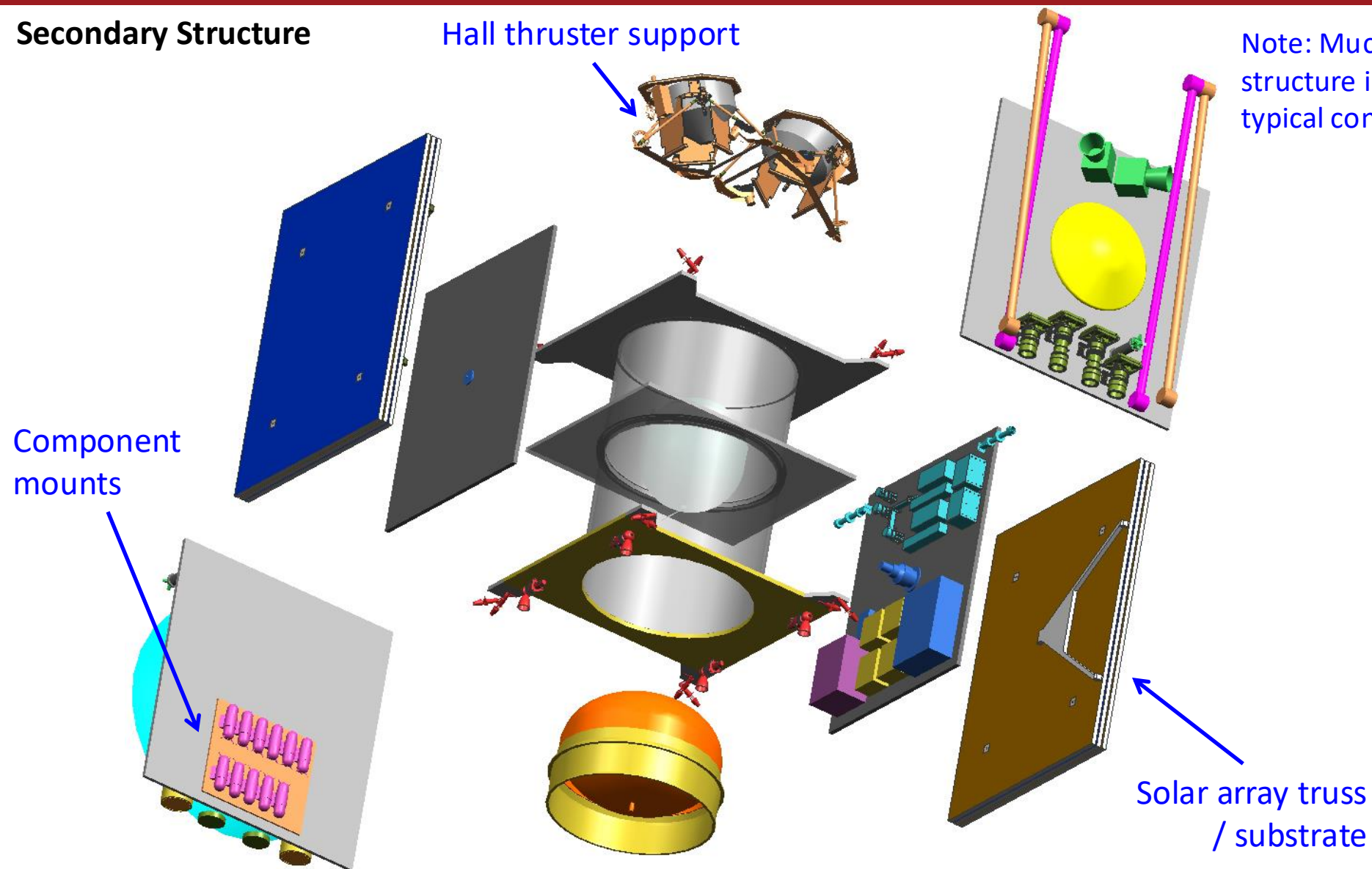
• Trade Studies

Analysis and Test

Design Process

Case Studies

• Secondary Structure



Example: Asteroid Sample Return Mission

Mechanical

Introduction

• Overview

• **Examples**

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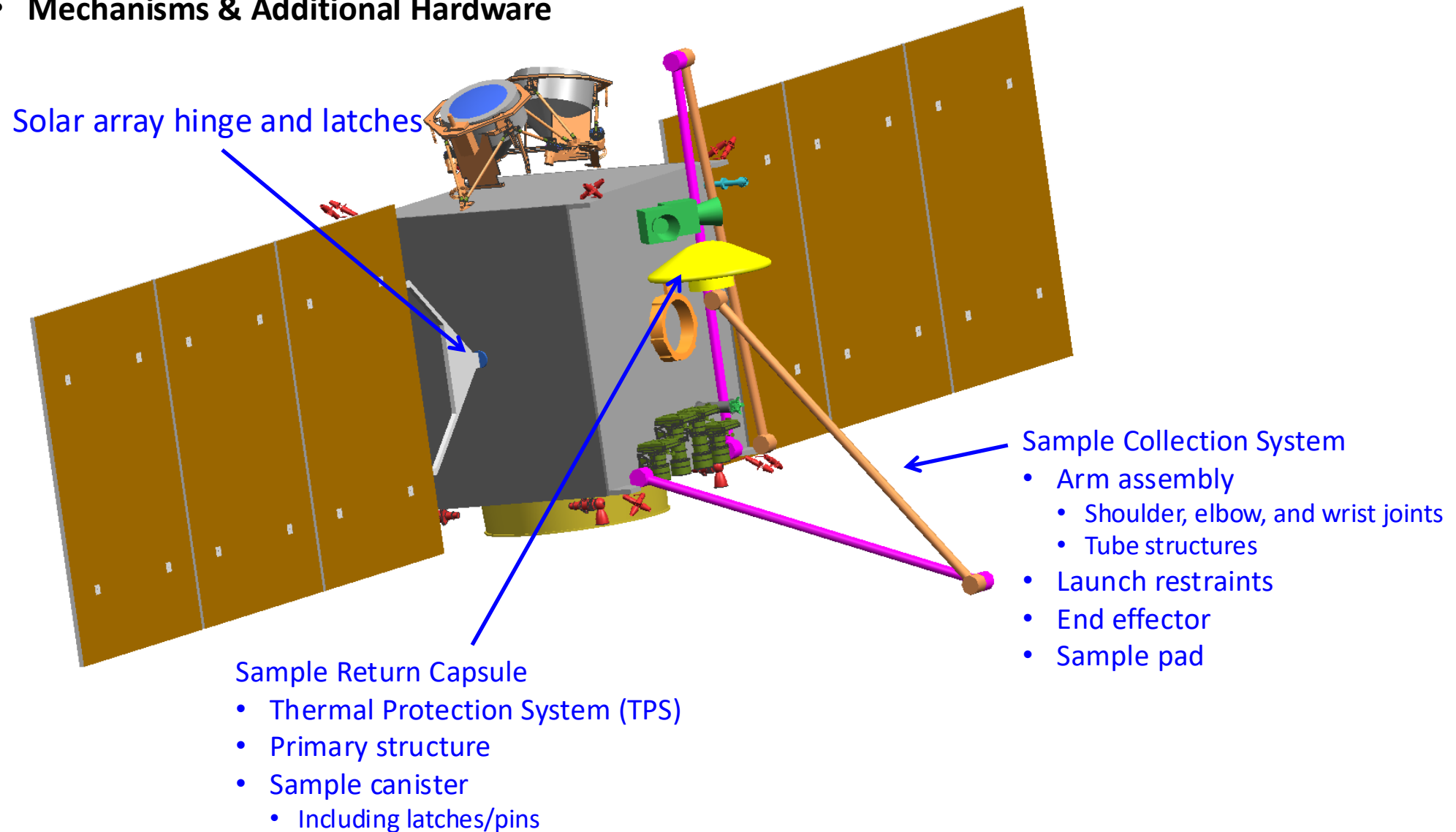
• Trade Studies

Analysis and Test

Design Process

Case Studies

• Mechanisms & Additional Hardware



Example: Asteroid Sample Return Mission

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• Overview

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• Mass Estimation

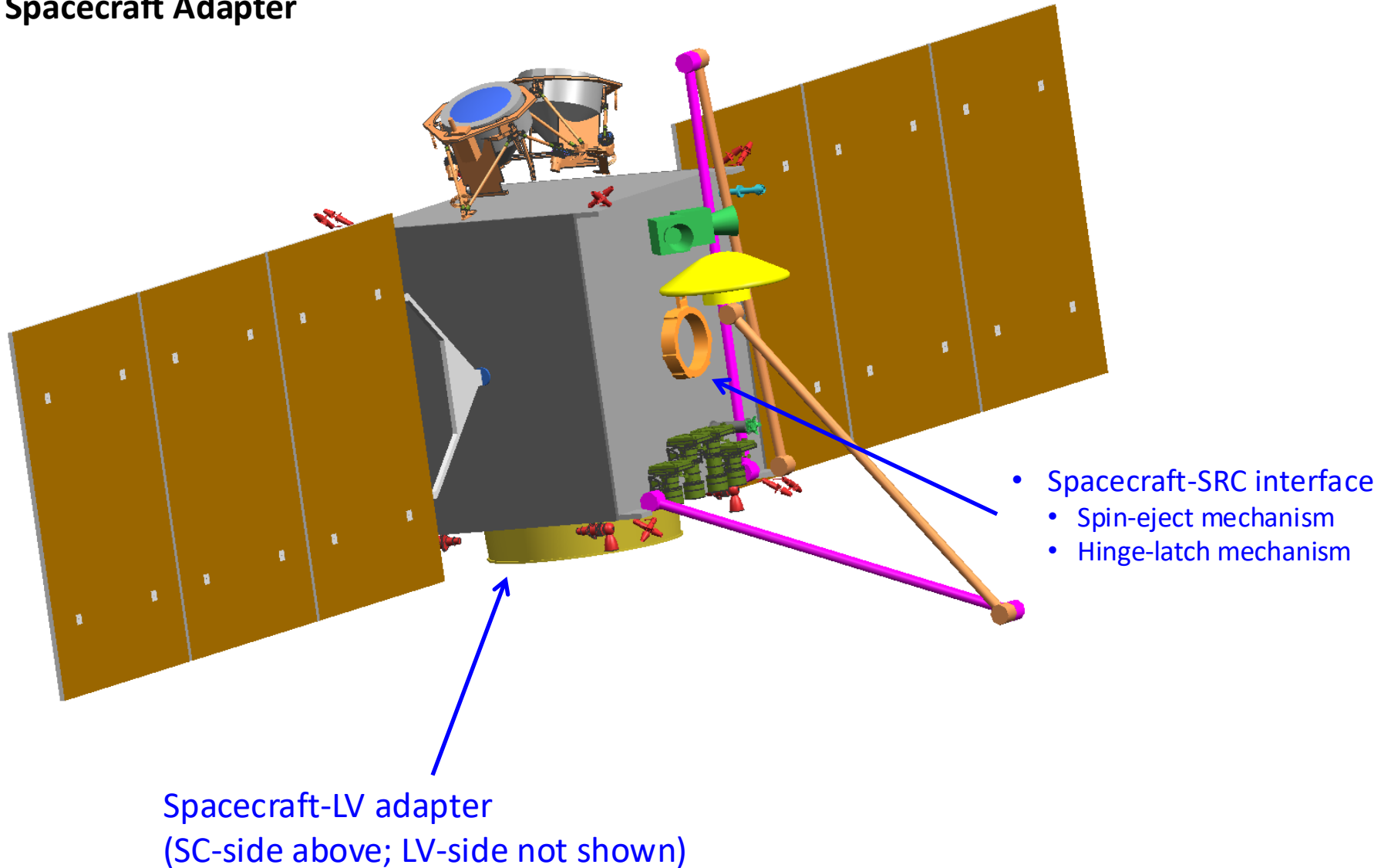
• Trade Studies

Analysis and Test

Design Process

Case Studies

• Spacecraft Adapter



Example: Asteroid Sample Return Mission

Mechanical

Introduction

• Overview

• Examples

Constraints & Reqs.

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• Mechanisms

• Configuration

• Loads and

Environments

• Mass Estimation

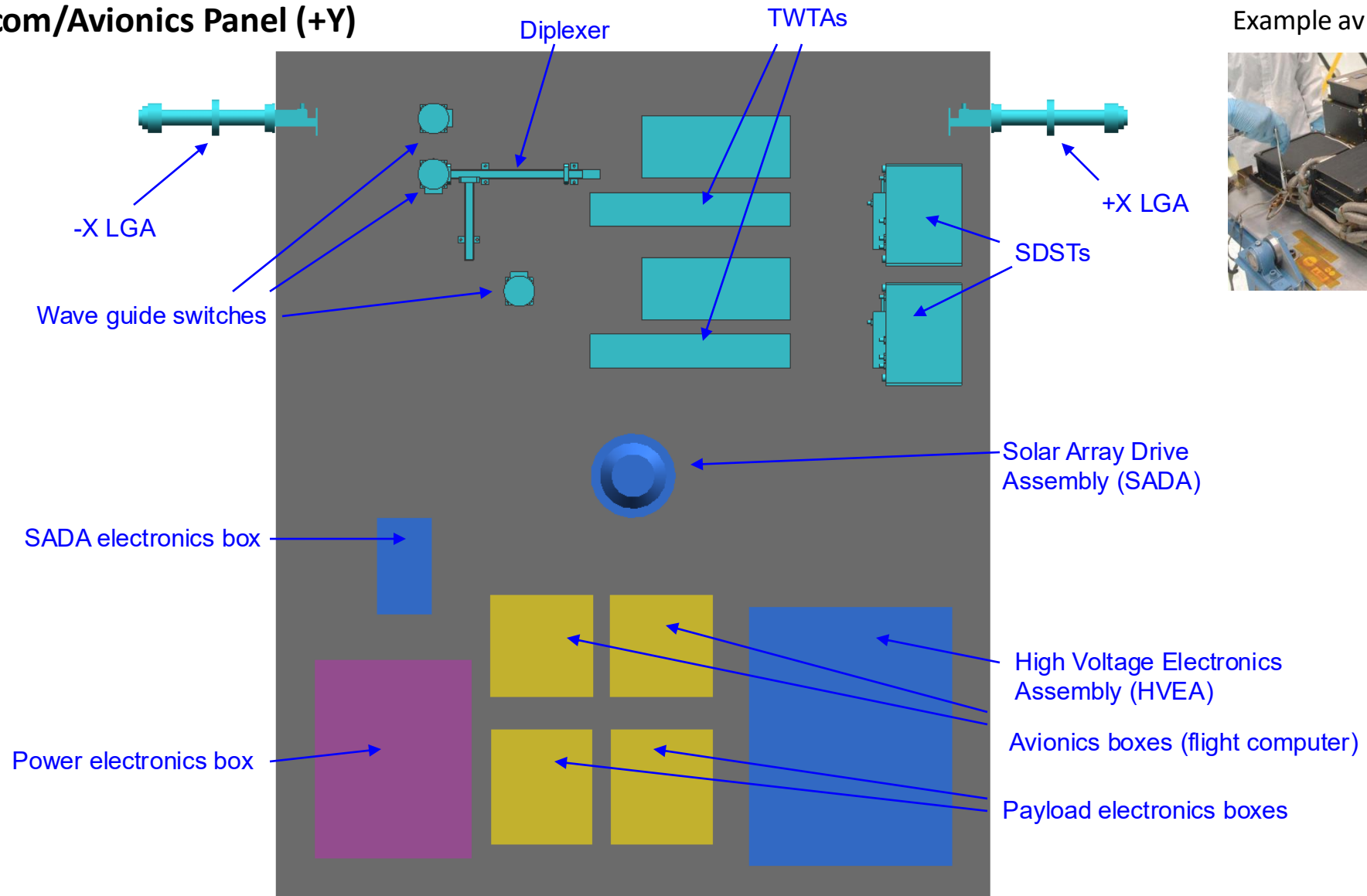
• Trade Studies

Analysis and Test

Design Process

Case Studies

• Telecom/Avionics Panel (+Y)



Example avionics from Hubble



Example: Asteroid Sample Return Mission

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Environments

• Mass Estimation

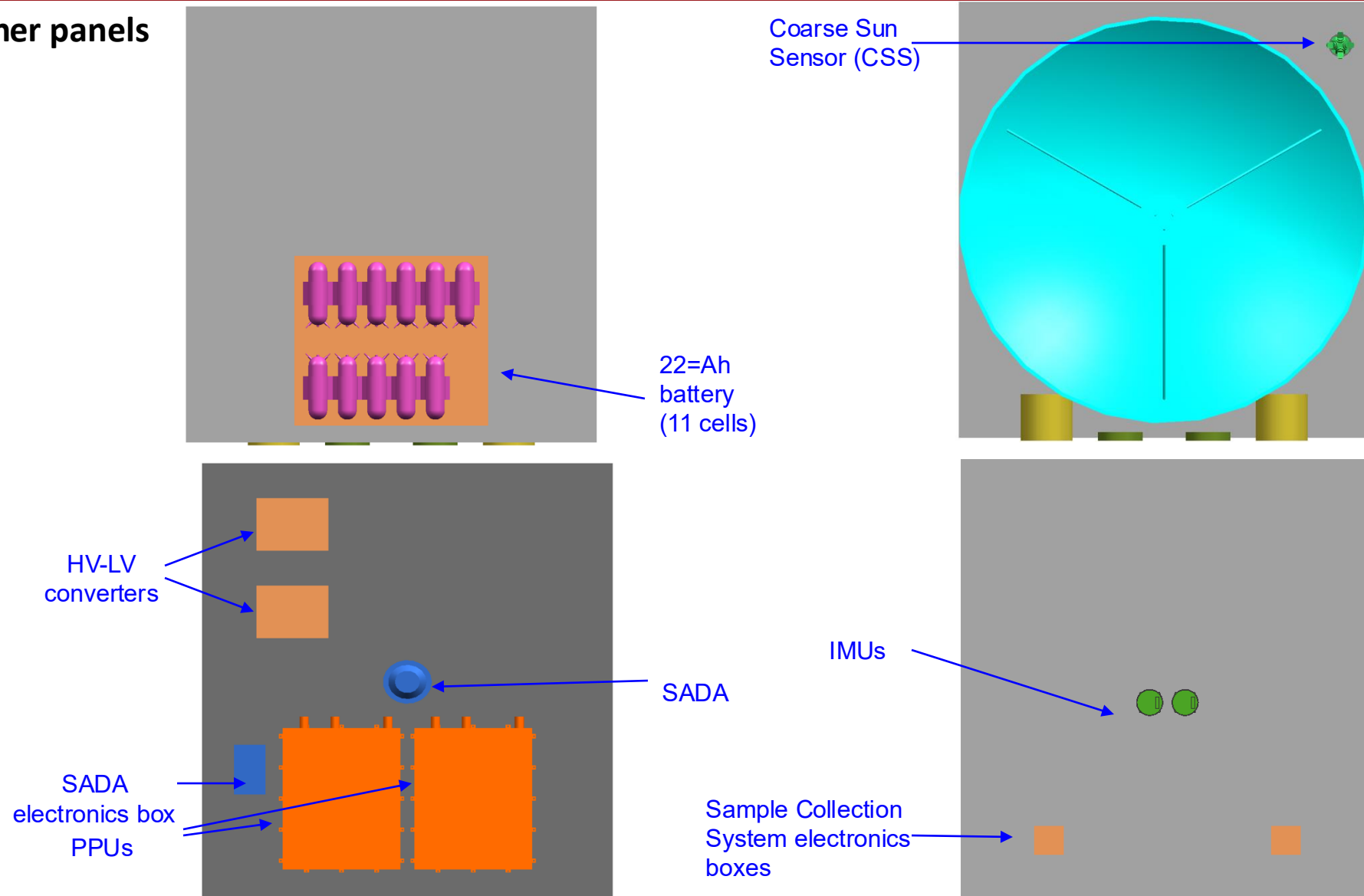
• Trade Studies

Analysis and Test

Design Process

Case Studies

• Other panels



Example: Asteroid Sample Return Mission

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• Mass Estimation

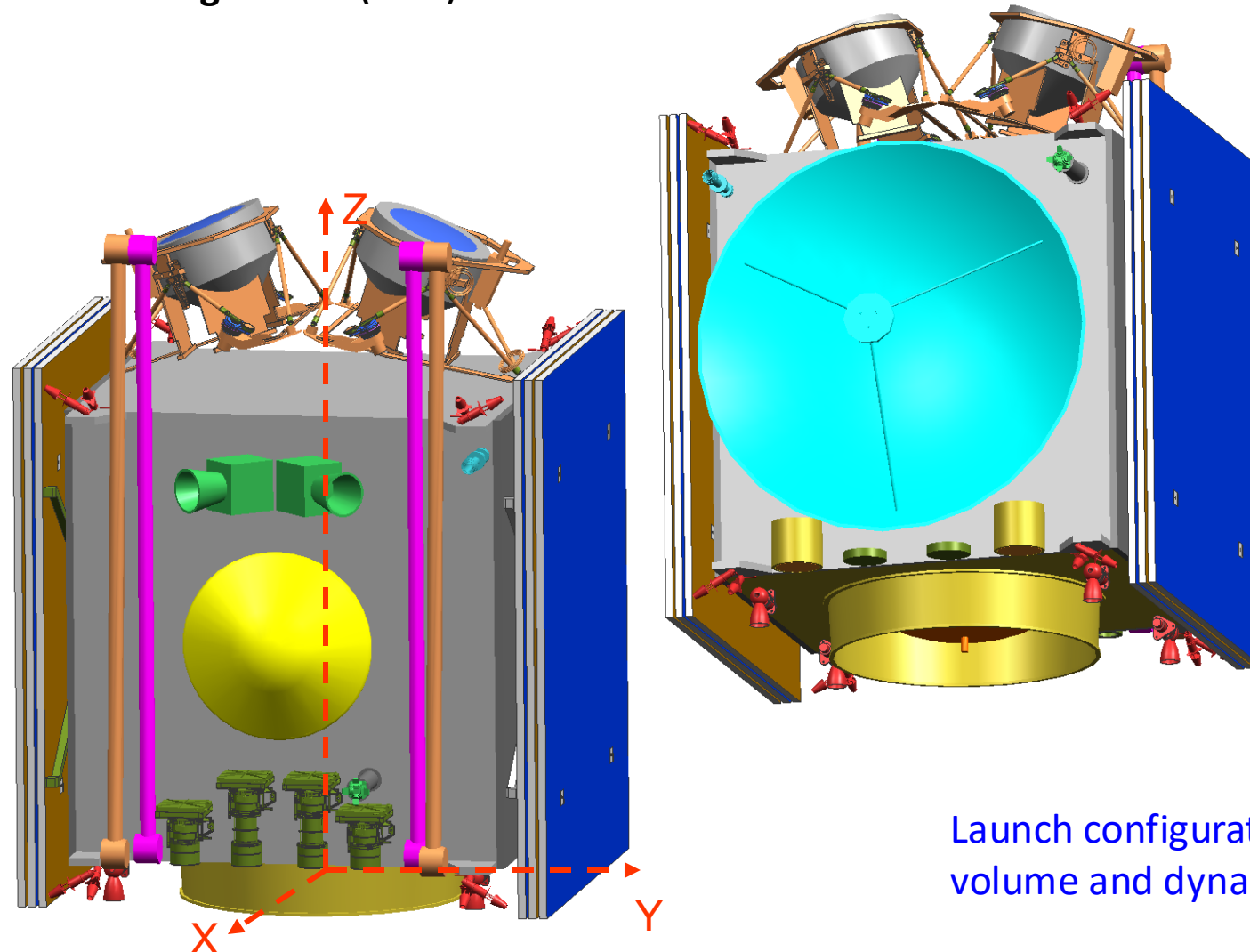
• Trade Studies

Analysis and Test

Design Process

Case Studies

• Spacecraft Launch Configuration (1 of 2)



Launch configuration is critical for both volume and dynamics constraints

Example: Asteroid Sample Return Mission

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Introduction

• Overview

• Examples

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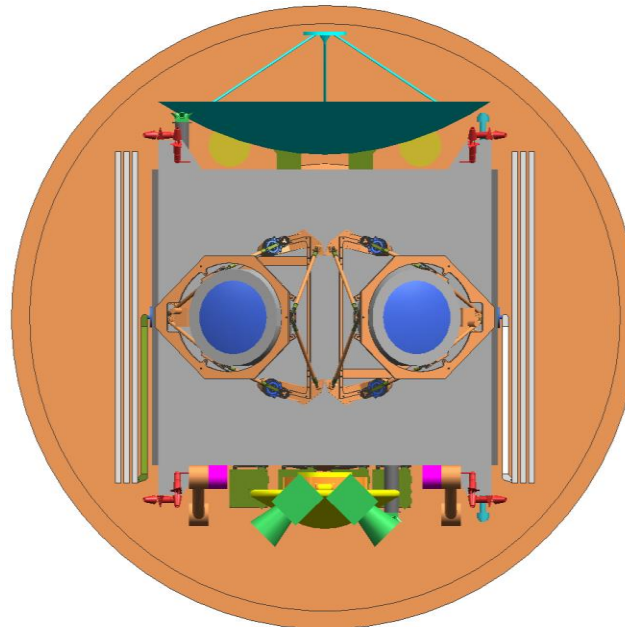
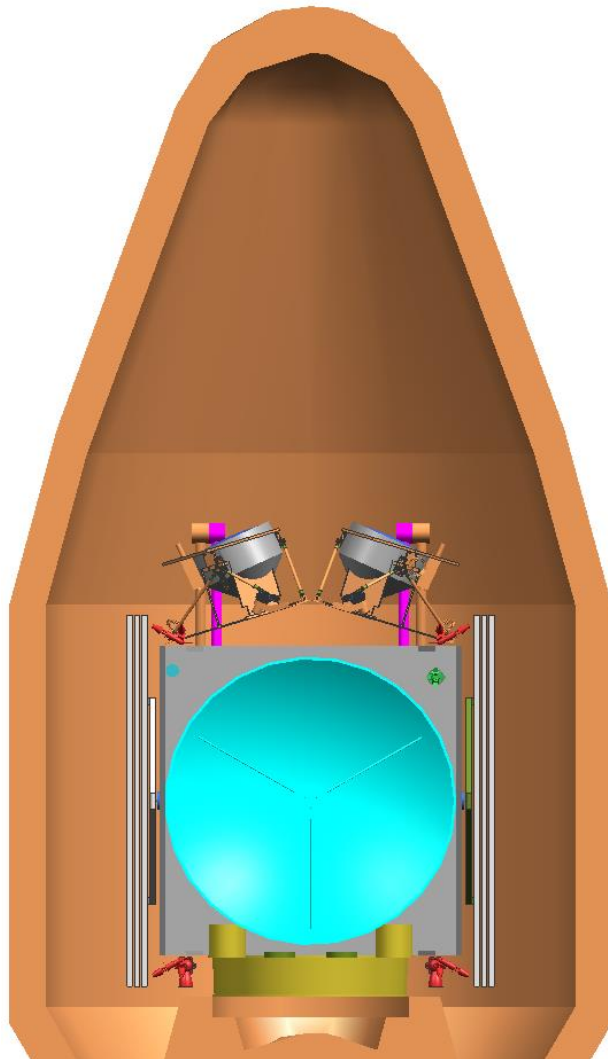
• Trade Studies

Analysis and Test

Design Process

Case Studies

• Spacecraft Launch Configuration (2 of 2)



Minimum clearance ensures no contact throughout launch

• Transportation to the Launch Site



Design also needs to consider loads due to transportation (e.g., spacecraft might be on its side)

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and
Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

Mechanical Subsystem Constraints and Requirements

Mechanical Subsystem Constraints and Drivers

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

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- Structures
- Materials
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Analysis and Test

Design Process

Case Studies

• Constraints

- Cost/budget
- Development and integration schedule
- Selected launch vehicle profile and interfaces
 - Including launch environment, mass, adapter interface
- Manufacturability
- Testability
 - Including access needs, Earth-gravity loads, mechanical ground support equipment (MGSE)



• Drivers

- Mission performance specifications
 - Including duration, trajectory, orbiter vs lander/rover
- Deployments and mechanisms
 - Including solar arrays, antennas, booms
- Payload accommodations
 - Including pointing, FOVs, alignments
- Subsystem accommodations
 - Including FOVs for ACS sensors and thrusters, alignments for ACS sensors and thrusters, harness routing, thermal pathways, KOZs
- Space environment
 - Including thermal, radiation, vacuum (e.g., outgassing – when trapped materials are released in vacuum)
- Mass & mass properties

System to Mechanical Requirements Flowdown

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
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Analysis and Test

Design Process

Case Studies

- **System requirements start from the mission objectives, but are heavily influenced by factors like cost and schedule**
- Requirements flow to the mechanical subsystem to ensure the spacecraft and payload can perform their functions (e.g., trajectory, pointing)
- General process:
 1. Science/Mission requirements
 2. Spacecraft system requirements
 3. Mechanical subsystem requirements
 4. Spacecraft conceptual design
 5. Structural/thermal analysis
 6. Revisit requirements until satisfactory
 7. Spacecraft detailed design
 8. Iterate 5 – 7 as necessary

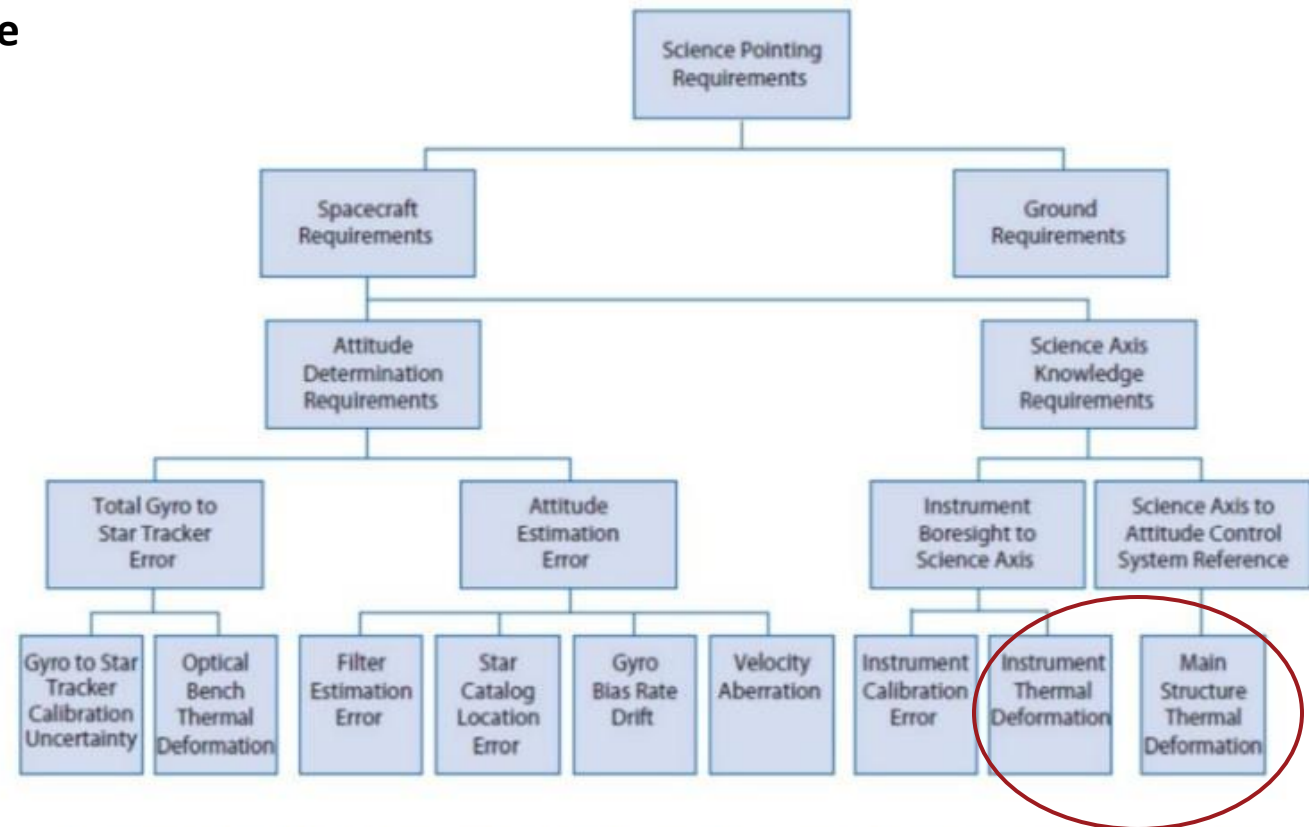


Figure 4.2-5 Allocation and Flowdown of Science Pointing Requirements

Sources: [1] <https://pressbooks-dev.oer.hawaii.edu/epet302/chapter/2-3/>

Common Mechanical/Structural Requirements

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

Requirement	Definition	Source
Strength	Load required to break, buckle or unacceptably deform a structure.	A fundamental structural requirement.
Structural Response	Magnitude and duration of vibration.	Prevent damage to components or launch vehicle.
Structural Life	The number of load cycles a structure can withstand before fatigue failure, or the time before creep or stress corrosion cracking cause failure.	A fundamental structural requirement.
Natural Frequency	The frequency at which a structure vibrates when excited by a load. Goal is to avoid exciting a structure at any of its natural frequencies to prevent large responses.	Launch: prevent interaction with launch vehicle. On-orbit: prevent interaction with ACS.
Mass Properties	Mass, CG, Moments and Products of Inertia.	Launch: limits of launch vehicles capabilities. On-orbit: limits of ACS and desired system dynamics.
Damping	A measure of how a structure dissipates vibration energy.	Used to limit the structural response.
Stiffness	Amount of load required to produce a unit deflection.	Derived for substructures to make sure they are stable and have adequate natural frequencies.
Dynamic Envelope	The volume that the vibrating spacecraft must not violate during launch in side the rocket fairing.	Launch Vehicle fairing effective size during launch.
Positional Stability	A measure of the ability to keep a certain location or direction. Yielding, thermal distortions, creep, vibrations and joint slip are all detrimental to positional stability.	Imposed to make sure sensitive instruments have proper orientation during the mission.
Mechanical Interface	Bolt patterns or feature shapes that control how 2 or more components will be joined.	Derived to minimize loads and ease of assembly.

Typically derived from...

Max expected loads during launch, landing, etc

Max expected loads during launch, landing, etc

Cycling due to number of articulations or other repetitive events

Launch vehicle frequencies, ACS frequencies

Launch vehicle and ACS stability needs

Payload stability needs

Strength, frequency, and deflection requirements

Launch vehicle fairing

Payload and ACS sensor needs

Subsystem needs, testability needs

Source: Spacecraft Structures and Mechanisms: From Concept to Launch by Thomas P. Sarafin (ed)., Space Technology Library, 1995 (4th printing 2003)

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and
Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

Mechanical Subsystem Design

Mechanical Subsystem Design

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies
- Analysis and Test
- Design Process
- Case Studies

• Primary Structure

- Houses majority of spacecraft components
- Transmits loads through the spacecraft to the interface of the launch vehicle
- Provides attachment points for payloads and associated components
- Typically core structure with subsystem-specific panels (e.g., avionics, telecom)
 - Note that panels are often integrated and tested individually before assembly
- Sized to accommodate propulsion subsystem and overall number of electronics boxes

• Secondary Structure

- Bridges the primary structure and provides attachments/mounts to components
- For example, solar array truss/substrate, equipment benches, etc.

• Tertiary Structure

- Additional load support and/or attachment hardware such as brackets

• Mechanisms

- Devices that move, rotate, slide, or separate
- Examples include release devices, deployables (hinges, booms), motors, gimbals, etc.

• Additional Hardware

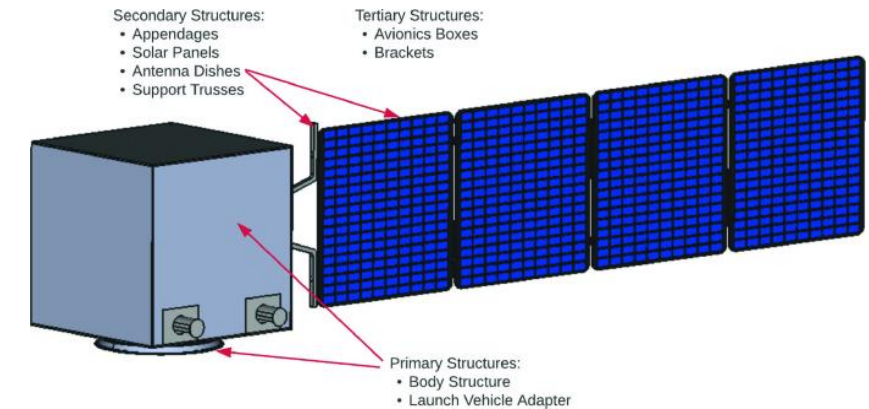
- Ballast mass (for mass properties), payload attachments, telescope barrels, etc.
- Other items, such as Sample Return Capsules are often owned by the mechanical team

• LV & Spacecraft Adapters

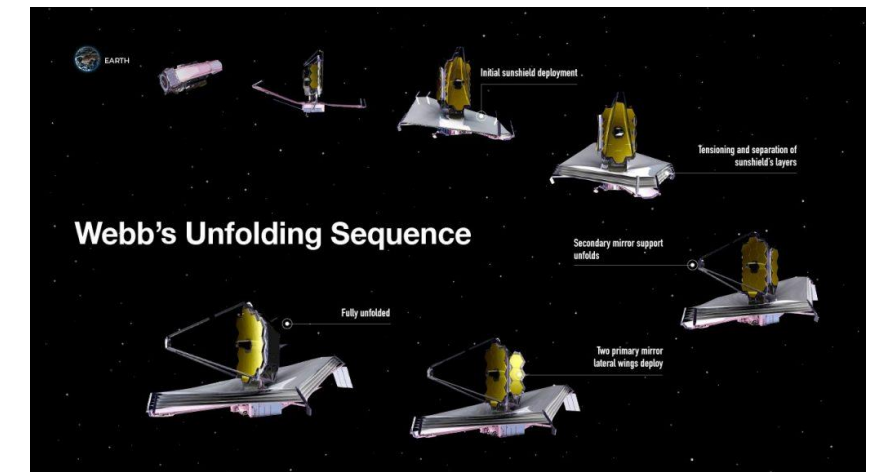
- Interfaces between LV and spacecraft, multiple spacecraft, probes, etc.

• Harness

- All of the cables/wires that are required to provide power & data



Spacecraft Structure Types



JWST Deployable Sunshield (NASA)

Common Structures

Mechanical

Introduction

• Overview

• Examples

Constraints & Reqs.

Design

• Structures

• Materials

• Mechanisms

• Configuration

• Loads and

Environments

• Mass Estimation

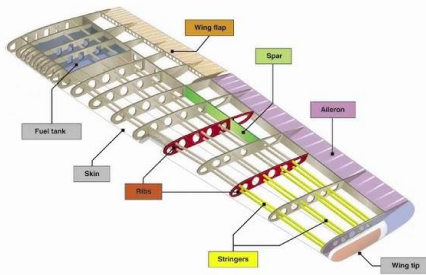
• Trade Studies

Analysis and Test

Design Process

Case Studies

- Some common structures (can overlap) are shown below:



Aircraft wing with stringers

Skin Frame Structures

- Common across aerospace industry, especially aviation
- Stringers on structure carry axial load and skin carries shear and axial loads



Sandwich panel (SatSearch)

- Metal/composite sheets with foam/honeycomb core
- Increased thickness for increased stiffness, at cost of small mass increase (bending/buckling/vibration)
- More failure modes such as through adhesive

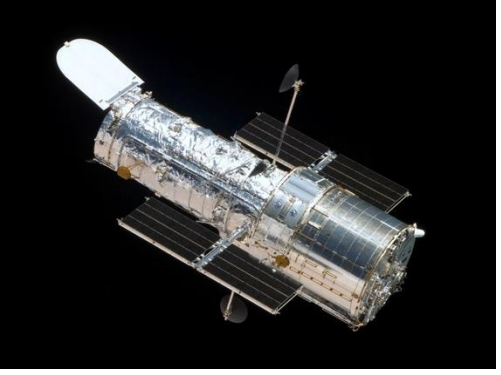
Sandwich Structures



Roman Space Telescope primary structure (NASA)

- Isogrids are mass efficient structures with cut-outs
- They are costly, complex, and difficult to machine
- Can be metal or composite (requires intricate mold)

Isogrids



Hubble Space Telescope (NASA)

- Can be monocoque (cylindrical skin) or semi-monocoque/stiffened cylinders (using skin-stringer)
- Typical for fuselages such as Space Shuttle
- Used when access to interior is important

Cylinder Structures

Materials Selection

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- **Materials**
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

- **Materials selection must ensure sufficient robustness to loads (e.g., launch) and environments (e.g., thermal, radiation)**
- **Material properties of interest**
 - Density, yield strength (plastic deformation rather than elastic), ultimate strength, strength-to-weight ratio, elastic coefficient (stiffness), coefficient of thermal expansion (CTE), thermal conductivity, radiation resistance
- **Materials**
 - Aluminum alloys are most common because of high stiffness/strength to weight ratio
 - Steel is high strength, better for high temperature use, heavier
 - Titanium is high strength, resists corrosion (e.g., due to atomic oxygen, thermal, UV radiation), heavier
 - Polymers (e.g., Kapton, epoxy) are lighter, much weaker – also keep in mind glass transition temperature, uncommon
 - Ceramics are heavier, brittle, much better for high temperatures (some TPS), uncommon
 - Composites are strong and much lighter – expensive, material testing and reliability issues, complex, difficult to join

Material	Density	Strength	Temp Tolerance	Applications	Unique Benefits	Unique Issues
Aluminum	Low	High	Medium	Panels, framework	Ease of manufacture	-
Steel	High	High	High	Bolts, fasteners	-	-
Titanium	Med-High	High	Medium	Brackets, fuel tanks, joints	Corrosion resistance	-
Polymers	Low	Low	Low	Insulation, seals	Ductility	Glass temperature
Ceramics	Medium	Medium	High	Coatings, TPS	Thermal properties	Brittle
Composites	Low	High	Med-High	Panels, booms, trusses	Optimizable properties	Expensive and complex

Materials Selection: Composites

Mechanical

Introduction

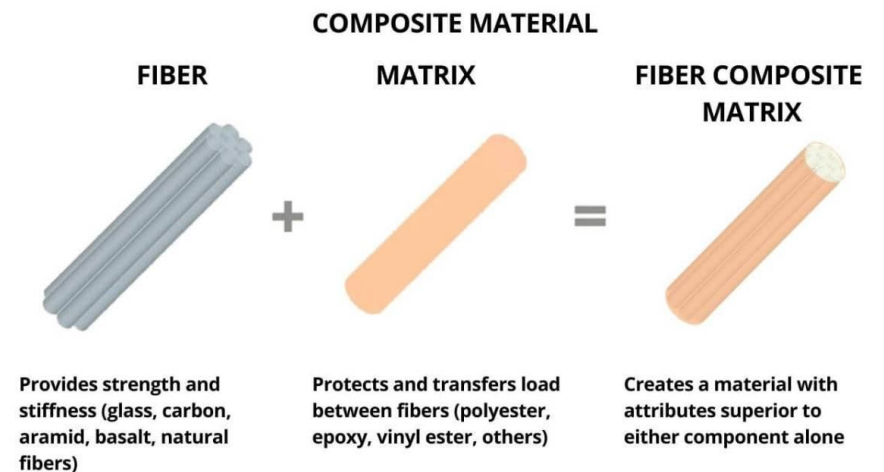
- Overview
- Examples

Constraints & Reqs.

Design

- Structures
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- Mechanisms
- Configuration
- Loads and Environments
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- Analysis and Test
- Design Process
- Case Studies

- **Made up of fiber and matrix**
 - Multiple layers of often varying fiber orientations
 - Combination creates a material with properties stronger than either material alone
- **Are orthotropic**
 - Different mechanical characteristics along each axis
- **Require expensive and extensive manufacturing and testing because of manufacture variations**
- **Fibers:**
 - Carbon fiber
 - Glass fiber
 - Aramid fiber (a type of synthetic fiber)
- **Polymer Matrices:**
 - Epoxy
 - Cyanate esters (a type of resin)
- **Failure is direction dependent**
 - Many failure modes exist and need to be considered when selecting a composite
- **Fiber/matrix selection & ply angle layup are main design parameters of laminates**



Mechanisms and Deployables

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

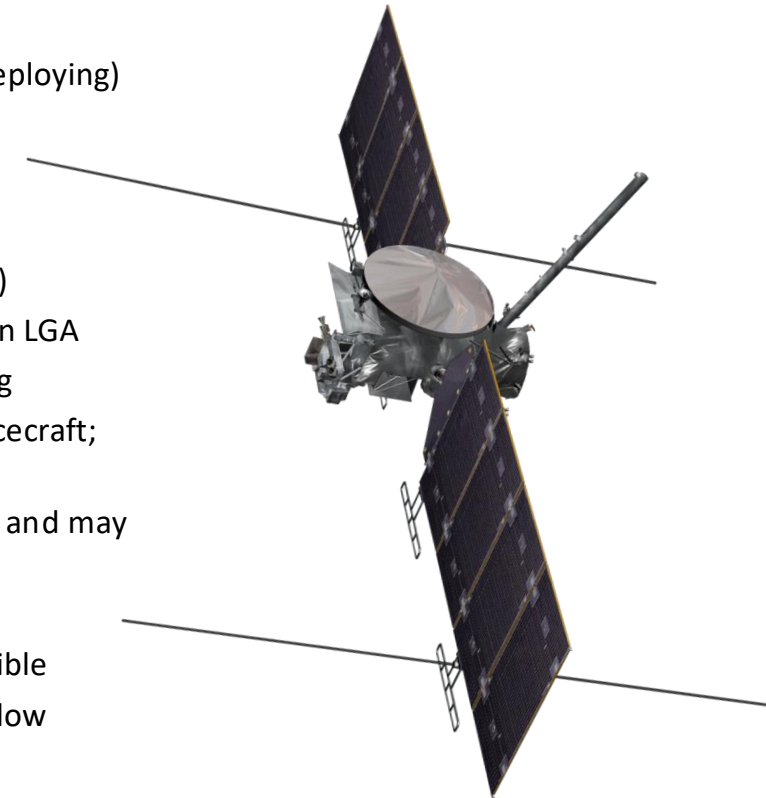
- Structures
- Materials
- **Mechanisms**
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

- **Mechanisms**
 - **Actuators:** Enable movement in systems (e.g., motors, servos), especially common in robotics
 - **Gimbals:** Control orientation of instruments, antennas, or solar arrays (a type of actuator)
 - **Hinges:** Facilitate rotational movement between spacecraft parts (e.g., solar array panels when deploying)
 - **Restraints and locks:** Secure deployables during launch and operation
- **Deployables**
 - **Solar Arrays:** Stowed for launch, deployed immediately after LV separation (typically autonomous)
 - **Antennas:** HGA may be stowed for launch, typically deployed after communications established on LGA
 - **Radiators:** Large radiators may be stowed for launch, typically deployed after initial commissioning
 - **Booms:** Sensitive components (e.g., magnetometers) may be on booms to reduce noise from spacecraft; stowed for launch, typically deployed after initial commissioning
 - **Instruments:** Large and/or sensitive instruments may be stowed for launch (e.g., radars, cameras) and may also have covers, typically deployed after initial commissioning
 - **Deployables are actuated using:**
 - Pyros: Simple, fast; can be susceptible to misfire or early fire, create shock and debris, irreversible
 - Springs: Simple, reliable, can be reversible; force limited, sensitive to environment and wear, slow
 - Motors: Controlled, reusable, low shock, scalable; complex, heavy, high power and weight
- **Design challenges**
 - Ensuring reliability in extreme conditions (vacuum, temperature fluctuations)
 - Ensuring spacecraft robustness to deployment shock
 - Avoiding deployment failures and designing for redundancy



Europa clipper has deployable solar array, instrument boom, antennas, camera covers; it also has a gimballed solar array

Configuration Considerations (1 of 2)

• Payload accommodation requirements:

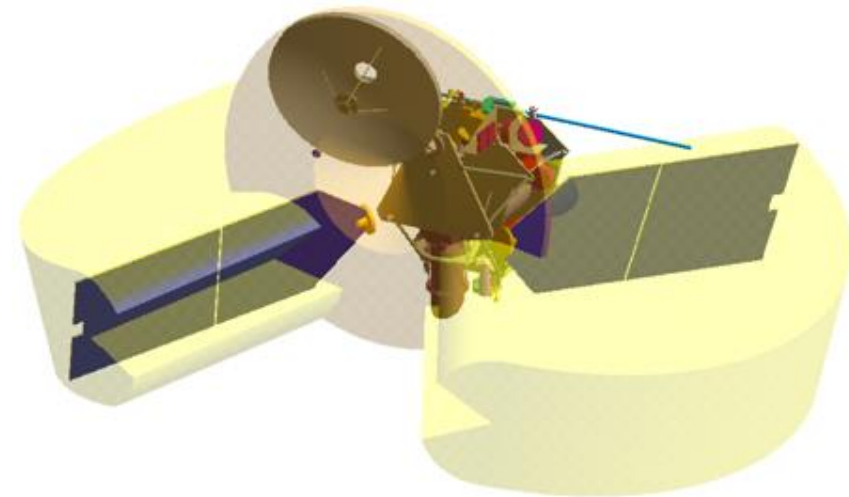
- Pointing, stability, alignment: Drives stiffness to minimize vibration response, mechanical/thermal deflections
- Fields of view (FOV) and keep out zones (KOZ): Drives configuration
- Electromagnetic interference (EMI): Drives configuration, separation of systems
- Special hardware such as masts, sample collection systems

• Mass property requirements:

- Driven by launch vehicle and ACS needs
 - Center of gravity (CG) is where weight is concentrated, affects stability
 - Moments of inertia (MOI) are how an object resists change in rotation, affects response to applied torque
 - Products of inertia (POI) are how mass is distributed along the principal axes, affects stability
- Symmetry is important
 - Solar arrays are typically in 2 or 4 panels configurations
 - Fuel tanks (due to mass) and wheels (due to control) are often centered at/near CG
 - Coupled thrusters split into clusters (often two strings)
 - Main engine typically on CG axis
- Ballast mass can be used to control mass properties

• Testability requirements:

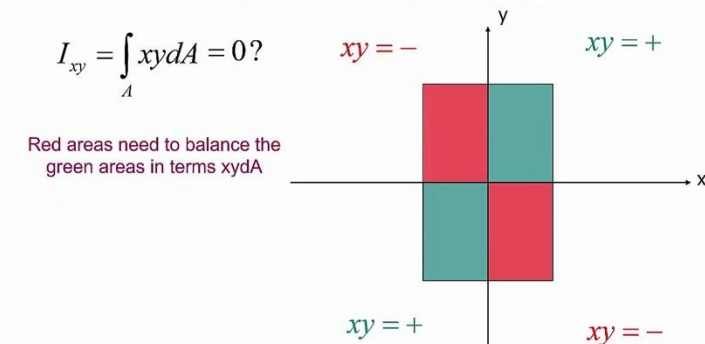
- Subsystem-specific panels help with integration flow (integrate/de-integrate single panels)



Spacecraft keep out zones

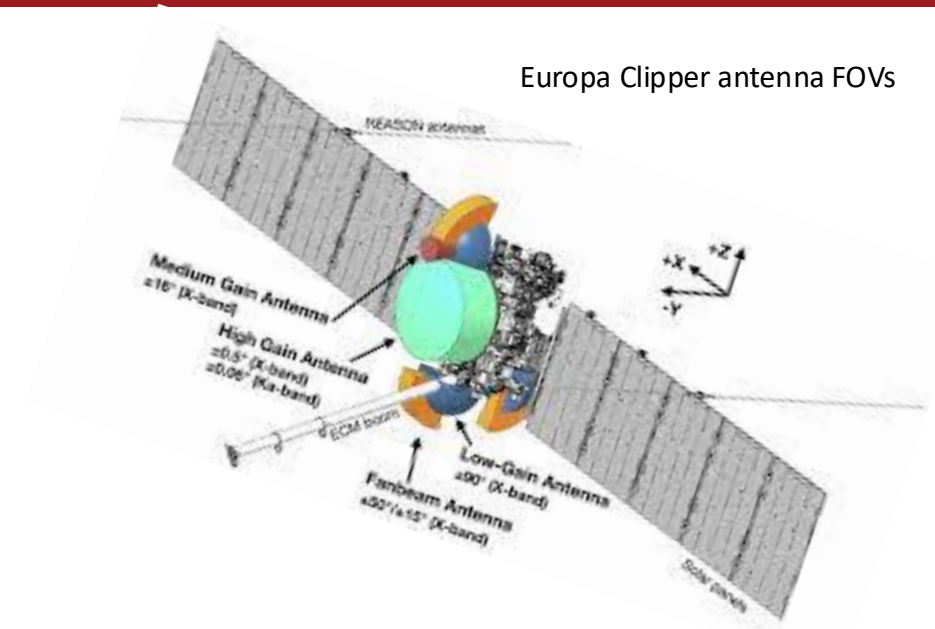
When is the Product Moment of Inertia Zero?

$$I_{xy} = \int_A xy dA = 0?$$

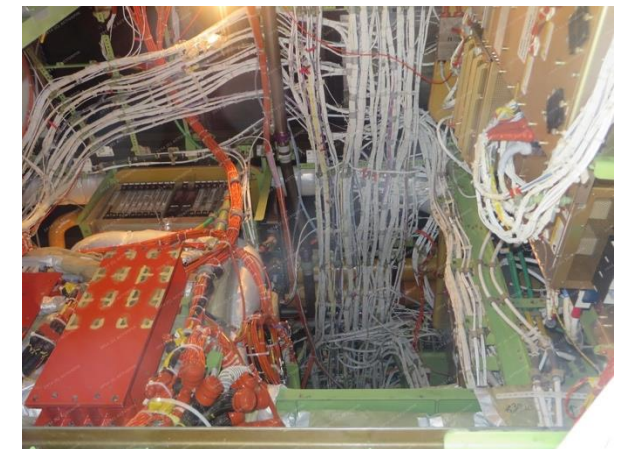


Configuration Considerations (2 of 2)

- **Propulsion requirements:**
 - Thrusters will have FOV requirements (for sufficient thrust) and KOZ requirements (for plume impingement)
 - Main engine should be close to CG
- **ACS requirements:**
 - Sensors will have similar requirements as Payload – pointing, stability, alignments, FOVs, KOZ, etc
 - Actuators should be close to CG
- **Avionics requirements:**
 - Electronics will have heat dissipation considerations
- **Telecom requirements:**
 - Antennas will have FOV and KOZ requirements
 - Electronics will have heat dissipation considerations
- **Power requirements:**
 - Electronics and RTGs will have heat dissipation considerations
- **Thermal requirements:**
 - Radiators will have FOV and KOZ requirements
 - Heat dissipation management will have configuration considerations
- **Harness requirements:**
 - Harness will require adequate mounting area and bend radius



Space shuttle harness routing



Loads and Environments: Overview

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

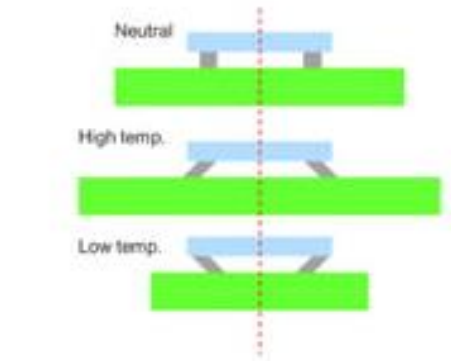
- Structures
- Materials
- Mechanisms
- Configuration
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Analysis and Test

Design Process

Case Studies

- **Mechanical subsystem has to survive loads encountered during launch and on-orbit operations**
 - Loads/accelerations, shock, vibrations, acoustics during launch, specified in LV user's guide ([SpaceX Falcon 9 User's Guide](#))
 - Shock from deployments (solar arrays, booms)
 - Vibrations from onboard equipment (reaction wheels, pumps)
 - Effects of micro-vibrations on sensitive instruments
- **Mechanical subsystem has to survive environments during launch and on-orbit operations**
 - Thermal
 - Expansion/contraction due to temperature variations, coefficient of thermal expansion (CTE) mismatch between two materials (builds up thermal stress)
 - Coefficient of thermal expansion (CTE): $\alpha = \frac{\Delta L}{L_0 \Delta T}$, where ΔL is change in length for a given ΔT , and L_0 is original length
 - Material fatigue under cyclic thermal stresses
 - Radiation
 - Radiation hardening and material selection for longevity
 - Mitigating radiation-induced degradation in composites and polymers
 - MMOD
 - Shielding strategies and materials
 - Impact testing and analysis



Effect of CTE Mismatch on Solder Joint Strain

Loads and Environments: Launch Accelerations

Mechanical

Introduction

- Overview
- Examples

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Analysis and Test

Design Process

Case Studies

• Launch accelerations

- Axial: Along spacecraft main axis
 - Due to upwards force of rocket, cause compression as spacecraft is pushed against payload adapter
 - Peak during early lift-off and periods of max thrust
- Lateral: Perpendicular to spacecraft main axis
 - Due to vibrations, wind, structural bending
 - Peak during Max Q (max dynamic pressure)

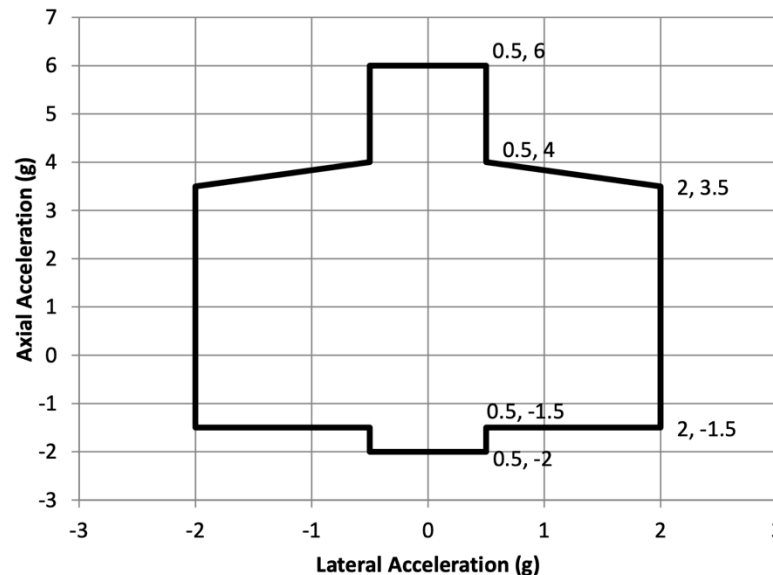
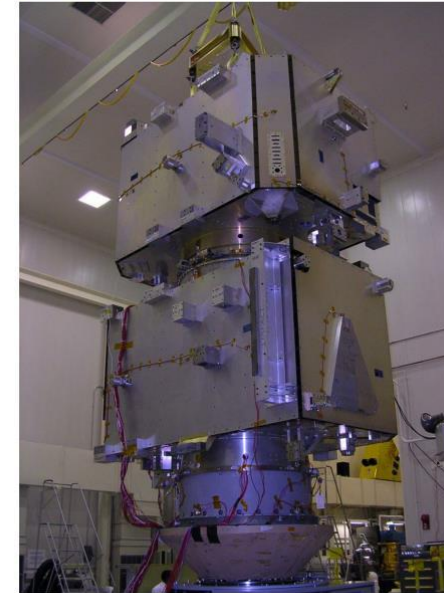


Figure 4-1: Falcon 9 and Falcon Heavy flight limit load factors for "standard" mass payloads (over 4,000 lb)



Spacecraft Stack – Dynamics Load Testing



STEREO - Solar Terrestrial Relations Observatory Mission

(ELR 4/29/05) - 4

STEREO spacecraft during load testing

Loads and Environments: Launch Vibrations

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
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- **Loads and Environments**

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Analysis and Test

Design Process

Case Studies

• Launch vibrations

- Due to operation of rocket engines and atmospheric turbulence
- Vary in frequency and amplitude, peak during ignition and liftoff
- Specified as sine or random
 - Sine: Helps to understand how structure behaves under specific frequencies, pinpoint resonances, most stressing
 - Specified as acceleration (g) at a given frequency
 - Random: Helps to understand behavior under non-deterministic and broad spectrum conditions of launch, most realistic
 - Specified as power spectral density (PSD, g^2/Hz), as well as RMS of acceleration (Grms) (square root of area under PSD curve)

Aspect	Sine Vibration	Random Vibration
Frequency	Single, controlled frequencies	Wide, simultaneous spectrum of frequencies
Pattern	Steady, repeating sine waves	Non-repeating, random variations
Real-World Causes	Engine combustion, structural bending modes, steady-state harmonics	Engine acoustic energy, atmospheric turbulence, high-frequency external vibrations
Primary Use	Simulates low-frequency, periodic oscillations and structural resonances	Simulates high-frequency, complex vibrational loads
Purpose	Identify resonant frequencies and structural weaknesses at specific frequencies	Evaluate overall durability against unpredictable, wide-range frequency vibrations
Testing Process	Uses pure, single-frequency sinusoidal waves applied at gradually increasing frequencies	Uses broad spectrum of frequencies simultaneously in a non-repeating, random pattern

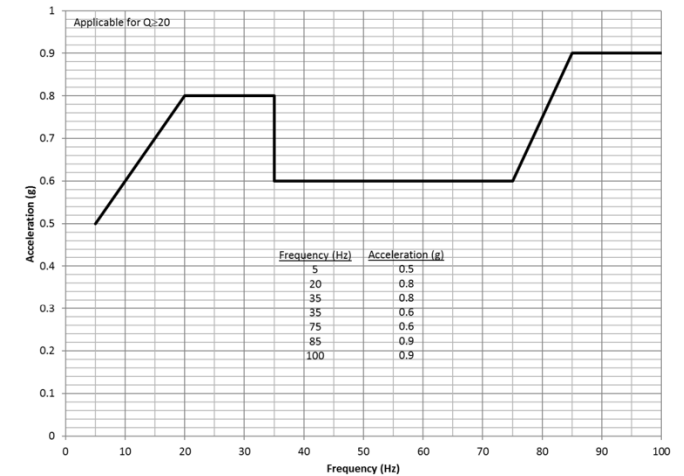


Figure 4-3: Maximum limit level axial equivalent sine environment for Falcon 9 and Falcon Heavy

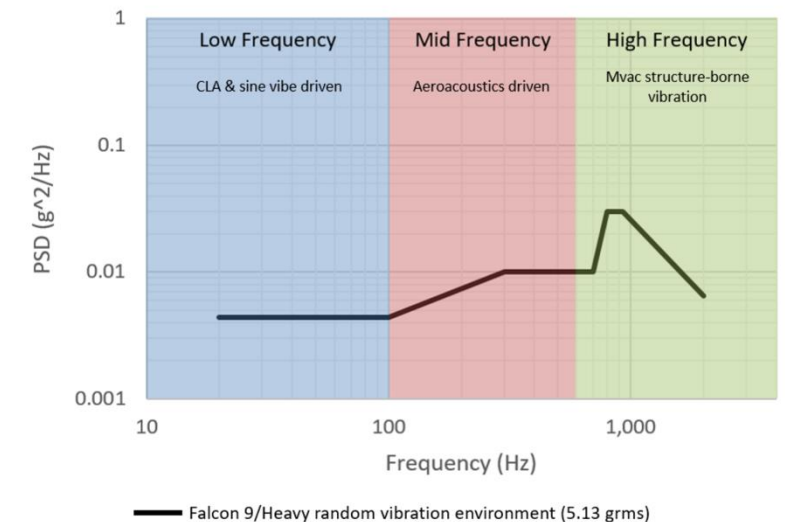


Figure 4-12: Falcon 9/Heavy frequency bin breakdown

Loads and Environments: Launch Acoustics and Shock

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration

• Loads and Environments

- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

• Launch acoustics

- High sound pressure levels from engines that reverberate through the launch vehicle structure
- Cause random vibrations that affect sensitive electronics
- Peak during liftoff
- Specified as sound pressure level (SPL) in dB as a function of frequency

• Launch shock

- Transient loads that occur during rapid events such as stage separations, engine shutdown, booster separation, fairing deployments
- Specified as shock response spectrum (SRS) in g

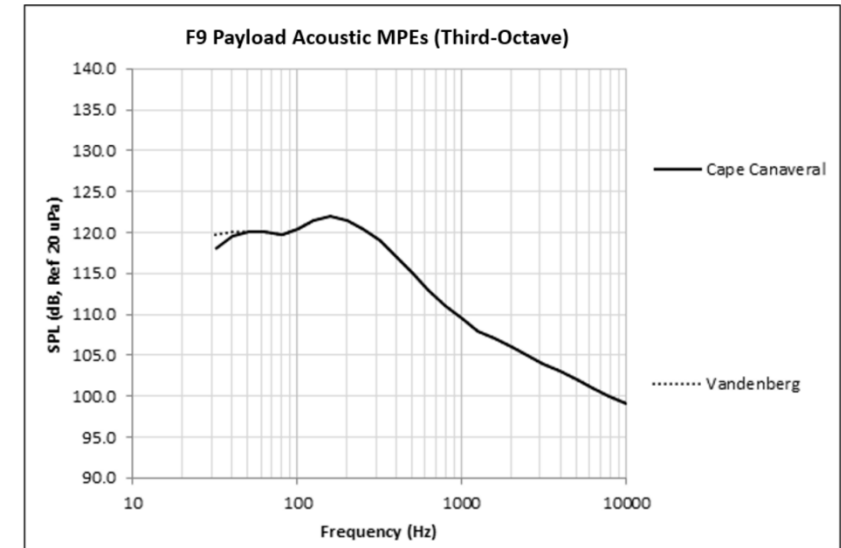


Figure 4-5: Falcon 9 maximum predicted acoustic environment (P95/50), 60% fill-factor, 131.3 dB OASPL (Cape Canaveral) and 131.4 dB OASPL (Vandenberg) in third octave, with fairing acoustic blankets

Table 4-9: Payload adapter-induced shock at the spacecraft separation plane (P95/50)

Frequency (Hz)	SRS (g)
100	30
1000	1,000
10000	1,000

Preliminary Mass Estimation

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- **Mass Estimation**
- Trade Studies

Analysis and Test

Design Process

Case Studies

- **Primary Structure**

- 5% of internal high-density components (electronics boxes, batteries, etc.)
- 15% of external high-density components (blankets, instruments, telecom, etc.)
- 20% of distributed components (dry tanks, feed system, thrusters, etc.)

- **Secondary Structure**

- 15% of primary structure (for junctions, stiffeners, brackets, etc.)
- 6% of externally supported systems (arrays, antennas, thrusters, etc.)
- 4% of propellant mass for strengtheners

- **Additional Items**

- 1.5 kg/m² for deployed panels, including solar arrays and radiators
- 1.0 kg each for latch/release systems
- 10% of primary structure for integration and interface hardware (shims, fasteners, etc.)
- 1% (3-axis) to 2.5% (spin-launched) of total dry mass for ballast
- 5% of total wet system mass for launch vehicle interface (75% on launcher, 25% on spacecraft)

- **Notes**

- Primary/secondary estimates are based on composites (add 12% if using aluminum)
- Reduce by 15-20% for optimized design and/or using advanced materials

While these are reasonable allocations, each organization/engineer will likely tailor these based on their experience and relevant heritage missions. The estimates will then be paired with increased contingency (20-30%) until they can be discarded in favor of actual CAD models with higher fidelity structural estimates.

Mechanical Design HW is based on this approach, which is typically used in conceptual design

Trade Study Analysis

Mechanical

Introduction

• Overview

• Examples

Constraints & Reqs.

Design

• Structures

• Materials

• Mechanisms

• Configuration

• Loads and Environments

• Mass Estimation

• Trade Studies

Analysis and Test

Design Process

Case Studies

- There may be several configurations that provide a working design, but consequently impact different subsystems or aspects of the mission
- Even if a configuration is chosen, different aspects of the mechanical design may be viable for alternate reasons
- **Trade studies** quantitatively compare alternative design choices with factors of varying weights
- Always compare what you **exchange** in a trade study
 - When optimizing mass properties, one might forgo configuration benefits or reduced margins for loads

Luxury Car Trade Study

Functional Requirement	Importance (1-4)	Lexus IS 300			BMW 330i			Mercedes C300		
		Value of Metric	Deg of Satisfaction (1-10)	Score	Value of Metric	Deg of Satisfaction (1-10)	Score	Value of Metric	Deg of Satisfaction (1-10)	Score
Purchase Cost	4	\$40,585	7	28	\$44,500	6.5	26	\$44,850	6.5	26
Annual Maintenance Cost	2	\$428	7.5	15	\$748	4	8	\$739	4	8
Horespower	2	241 hp	4	8	255 hp	4.5	9	255 hp	4.5	9
Fuel economy	3	31 mpg	6	18	34 mpg	7.5	22.5	36 mpg	8	24
Aesthetic	3	-	7	21	-	6	18	-	6	18
Sum Total				90			83.5			85
Best Option	Lexus IS 300									

Assumptions

1. I am an astronautical engineer with a starting salary of \$80,000, and no price should exceed my salary.
2. My driving environment will be commuting on freeway (for relevancy of fuel economy considerations).
3. Each car takes premium fuel (for fuel economy considerations).
4. Each car is up to date with current safety standards and technologies (auto-brake, sensors, etc.)
5. For degree of satisfaction, will round up to a multiple of 0.5 for cleanliness.

Common Mechanical/Structural Trade Studies

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation

• Trade Studies

Analysis and Test

Design Process

Case Studies

Trade	Point in Mission	Goals	See Chapters
Launch Vehicle	Launch	Evaluate how different LV lift capabilities, fairing sizes, and launch load environment impact payload.	3.1
S/C Configuration	Launch and On-Orbit	Position and arrange the payload and subsystems in the stowed and deployed configurations. Determine load paths for primary and secondary structures. Layout deployable hardware and estimate mass properties of components. Ensure that the stowed payload fits in the fairing.	14
Primary Structure	Launch	Use beam and mass model to derive design loads for various components. Investigate materials, payload shape, and attachment methods. Layout and size structural members for stiffness and strength.	15
Subsystem Support	Launch and On-Orbit	Show that antennas, instruments, and solar arrays have their required field of view. Ensure pointing and stability requirements are met for secondary and tertiary structures. Conceptual design and requirements for mechanisms. Decide if active or passive control of loads and responses is require.	14, 15.1, 19, 17
Mass Properties	Launch	Make sure structural subsystem and mechanisms meet mass allocation. Include uncertainty for new designs.	14 and 15

Source: Spacecraft Structures and Mechanisms: From Concept to Launch by Thomas P. Sarafin (ed)., Space Technology Library, 1995 (4th printing 2003)

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and
Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

Structural Analysis and Test

Structural Analysis and Testing

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

- **Structural analysis and testing determines if spacecraft can survive the mission and perform its necessary functions**
- **This is an iterative process**
 - If analysis and testing show a needed increase in mass, must update design and configuration accordingly, and then re-analyze
- **This ultimately leads to the validation and verification (V&V) of the subsystem and other system components**
- **Challenges in analysis**
 - Model accuracy and validation
 - Modeling of transient loads
 - Thermo-mechanical coupling
- **Challenges in testing**
 - Handling large and complex systems without damage
 - Simulating the launch environment on Earth
 - Testing for mission-specific scenarios (e.g., deep space, planetary landing)

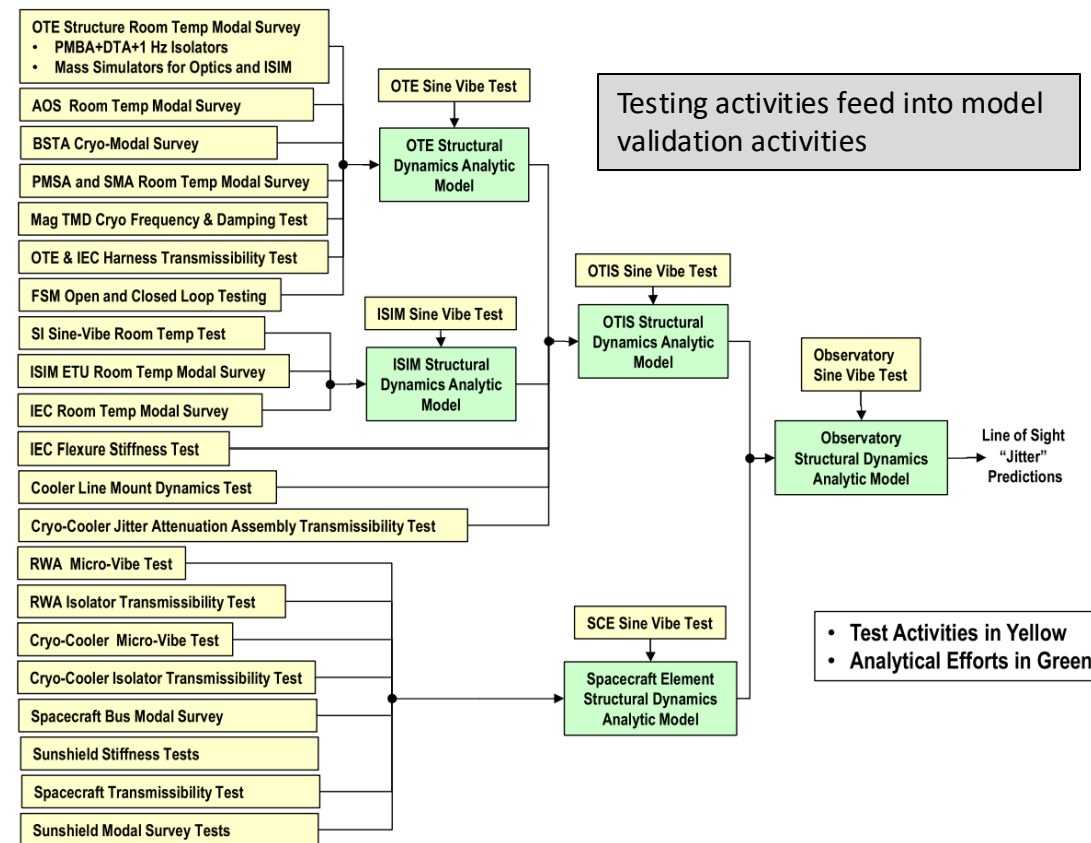


Figure 26. Verification of JWST jitter performance.

Structural Analysis and Testing

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

	Qualification Testing	Protoflight Testing	Acceptance Testing
Purpose	Verify that design meets all requirements with robust margins, identify weaknesses	Verify that design meets all requirements with sufficient margin, used when budget or schedule constraints limit number of test units	Verify that each flight unit (including duplicates or backup units) meets all requirements
Conducted on...	Prototype hardware not flown	Actual flight hardware that will be launched	Every flight unit to confirm functionality and manufacturing quality
Conditions	Testing levels exceed expected operational limits (~1.5 to 2x expected loads, vibrations, and temperature ranges)	Testing levels are slightly above operational limits but not as intense as qualification testing (often between qual and acceptance levels)	Testing levels are at or slightly below operational limits
Outcome	Confirms robustness of the design	Verifies that the flight hardware can withstand mission conditions without over-stressing it	Verifies that the unit can withstand mission conditions without over-stressing it, demonstrates functional readiness rather than robustness to extremes
Examples	Higher-level loads, vibration, shock, and thermal vacuum tests, to confirm flight readiness with robust margins for safety	Moderate-level loads, vibration, shock, and thermal vacuum tests, to confirm flight readiness with some margin for safety	Standard-level loads, vibration, shock, and thermal vacuum tests, to confirm flight readiness within expected mission parameters

Statics/Stress Analysis and Test

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

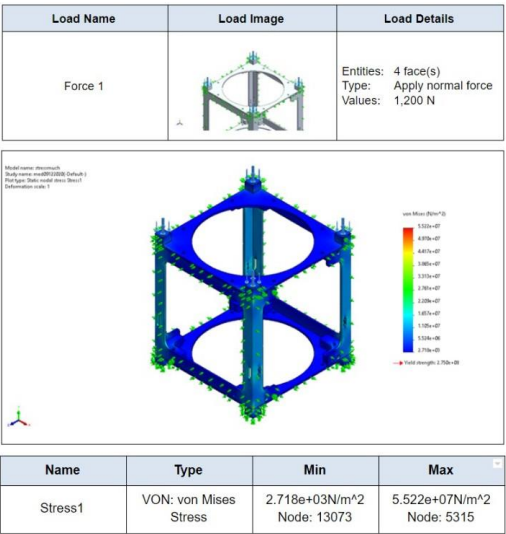
- Design Process
- Case Studies

• Stress analysis:

- Purpose: Determines stresses in structure and components and if they can withstand such loading; determine deflections and verify stiffness requirement
 - Coupled loads analysis (CLA) combines effects of loads originating from multiple sources/directions during launch
- Strategy: Finite Element Analysis (FEA) software simulates stresses, deformations, and vibrations under various load case
 - Examples: ANSYS, NASTRAN, Abaqus

• Static Load Testing:

- Purpose: Evaluates the spacecraft's ability to withstand static loads, simulating forces it will experience during launch
- Strategies:
 - Applies loads in a controlled manner, usually at or above expected load levels, to verify that structural integrity is maintained with robust margins
 - Quasi-static testing applies loads gradually and holds them steady for long durations, simulating scenarios such as launch



FEA for simple structure



OSIRES-REx on static load structure

Statics/Stress Analysis and Test

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

• Basic equations used for stress calculations

Concept	Equation & Application	Variables	Application
Normal Stress	$\sigma = F / A$	σ : Normal stress, F: Axial force A: Cross-sectional area	Stress perpendicular to a cross-sectional area due to axial loads. Important for launch load analysis.
Shear Stress	$\tau = V / A$	τ : Shear stress V: Shear force A: Cross-sectional area	Stress parallel to a cross-sectional area. Important for joints and fasteners subjected to shear during launch.
Strain	$\epsilon = \Delta L / L_0$ $\gamma = \Delta x / L$	ϵ : Normal strain γ : Shear strain ΔL : Change in length L_0 : Original length Δx : Lateral displacement L: Length over shear occurs	Deformation per unit length. Important for assessing structural elongation or angular distortion under load.
Hooke's Law	$\sigma = E * \epsilon$ $\tau = G * \gamma$	E: Young's modulus G: Shear modulus	Linear relationship between stress and strain in the elastic region. Important for deformation calculations.
Bending Stress	$\sigma = M * y / I$	σ : Bending stress M: Bending moment y : Distance from neutral axis I: Moment of inertia	Stress from bending moments in beams. Important for analyzing spacecraft booms, panels, and supports.
Torsional Shear Stress	$\tau = T * r / J$	τ : Shear stress T: Applied torque r: Radius J: Polar moment of inertia	Stress in rotating parts, like antennas or reaction wheels.

Statics/Stress Analysis and Test

Mechanical

Introduction

- Overview

- Examples

Constraints & Reqs.

Design

- Structures

- Materials

- Mechanisms

- Configuration

- Loads and

Environments

- Mass Estimation

- Trade Studies

Analysis and Test

Design Process

Case Studies

- Basic equations used for stress calculations

Concept	Equation & Application	Variables	Application
Thermal Stress	$\sigma_{\text{thermal}} = E * \alpha * \Delta T$	σ_{thermal} : Thermal stress E: Young's modulus α : Coefficient of thermal expansion ΔT : Temperature change	Stress induced by temperature changes. Important for spacecraft facing thermal cycles.
von Mises Stress	$\sigma_v = \sqrt{(\sigma_x^2 - \sigma_x \sigma_y + \sigma_y^2 + 3\tau_{xy}^2)}$	σ_v : von Mises stress σ_x, σ_y : Normal stresses τ_{xy} : Shear stress	Stress due to combined normal and shear stresses. Important for failure prediction under combined loading.
Safety Factor	SF = Material Strength / Applied Stress	SF: Safety factor Material Strength: Yield/Ultimate strength Applied Stress: Calculated stress	Ensures design stress remains within safe limits. Important for reliability under uncertain conditions. Yield: Typically 1.1 to 1.25 Ultimate: Typically 1.4 to 2
Axial Deformation	$\Delta L = F * L_o / (A * E)$	ΔL : Change in length F: Axial force L_o : Original length A: Cross-sectional area E: Young's modulus	Elongation or compression due to axial load. Important for assessing structural changes during launch acceleration.

Dynamics/Modal Analysis and Testing

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

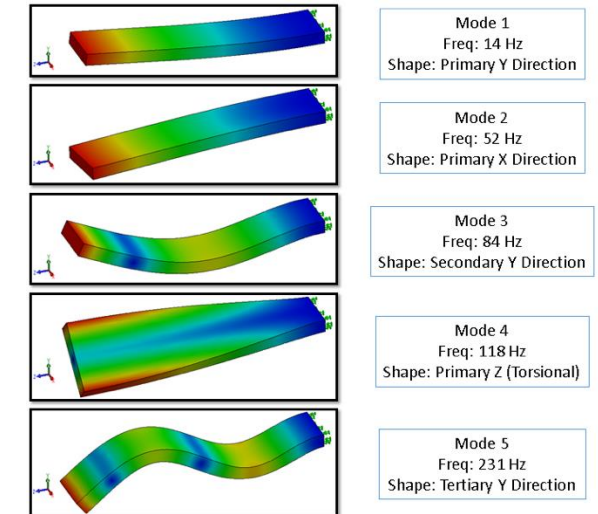
Case Studies

• Dynamics and modal analysis:

- Purpose: Determines natural frequencies, mode shapes, and damping; determines vibration loads
- Strategies:
 - Dynamic analysis assesses spacecraft under transient and time-dependent loads such as vibrations and shock
 - Examples: LS-DYNA, MSC ADAMS
 - Modal analysis assesses spacecraft natural frequencies to ensure resonance does not compromise structural integrity under launch and operational loads
 - Examples: HyperMesh, MSC NASTRAN

• Dynamics and modal testing:

- Vibration Testing
 - Purpose: Tests spacecraft's tolerance to vibrational environment of launch and separation events
 - Strategy: Shaker table subjects spacecraft to varying frequencies and amplitudes
 - Sinusoidal vibe testing concentrates energy on single frequency at a time to excite resonance, stressing
 - Random vibe testing covers many frequencies, closer to launch environment
- Shock Testing
 - Purpose: Tests spacecraft's tolerance to shock events like stage separation or pyro firings
 - Strategy: Impact device or explosive actuator generates sudden, high-energy shocks
- Acoustic Testing
 - Purpose: Tests spacecraft's tolerance to the noise environment of launch
 - Strategy: Acoustic chamber exposes the spacecraft to high-decibel sound waves
- Modal Testing
 - Purpose: Identifies the natural frequencies and vibration modes of the spacecraft structure
 - Strategy: Shaker table applies small oscillatory forces at different points to determine resonant frequencies



Modal analysis



Spacecraft acoustic testing

Thermal Analysis and Testing

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

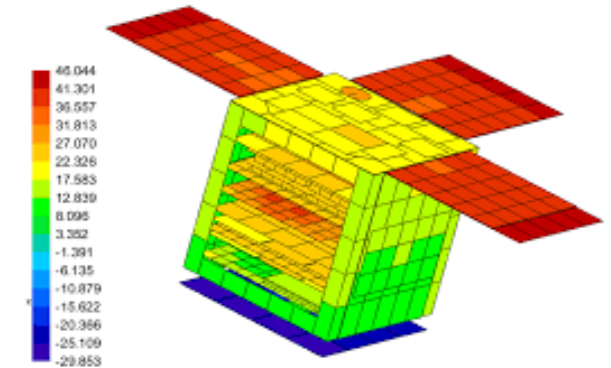
Case Studies

• Thermal analysis:

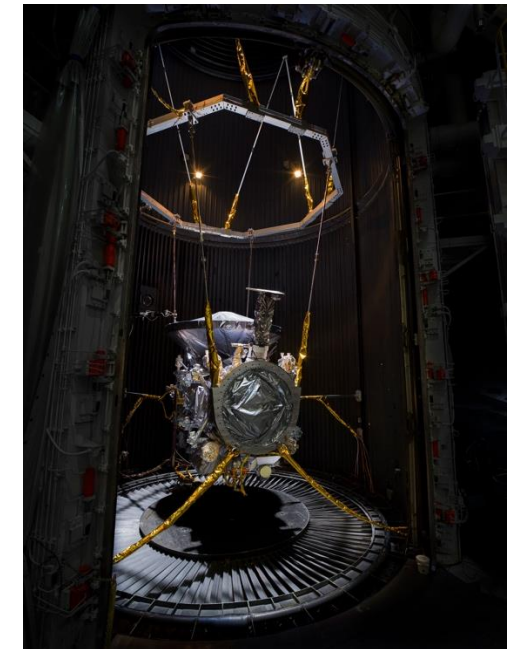
- Purpose: Determines robustness of spacecraft to operational and survival temperatures; determines stresses, strains, and deflections caused by temperature changes
- Strategy: Thermal analysis software analyzes temperature distributions and thermal stresses to predict thermal loads on spacecraft due to space environment
 - Examples: Thermal Desktop, Sinda

• Thermal testing:

- Thermal Vacuum Testing (TVAC)
 - Purpose: Ensures the spacecraft can withstand the extreme temperatures and vacuum conditions of space
 - Strategy: Places spacecraft in a thermal vacuum chamber, temperatures cycle and hold through extremes of mission profile
 - Types:
 - Thermal balance: Steady-state conditions to validate thermal models
 - Thermal cycle: Cycling conditions to test resilience to repeated temperature changes – e.g., due to repeated eclipses (can cause expansion, contraction, fatigue)
 - Thermal soak: Prolonged exposure to a single hot and cold temperature to test resilience to extremes
 - Thermal shock: Quick transitions between hot and cold temperatures to test resilience to rapid temperature change – e.g., due to eclipse exit



Spacecraft thermal analysis



Europa Clipper in TVac chamber

Fatigue Analysis and Testing

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

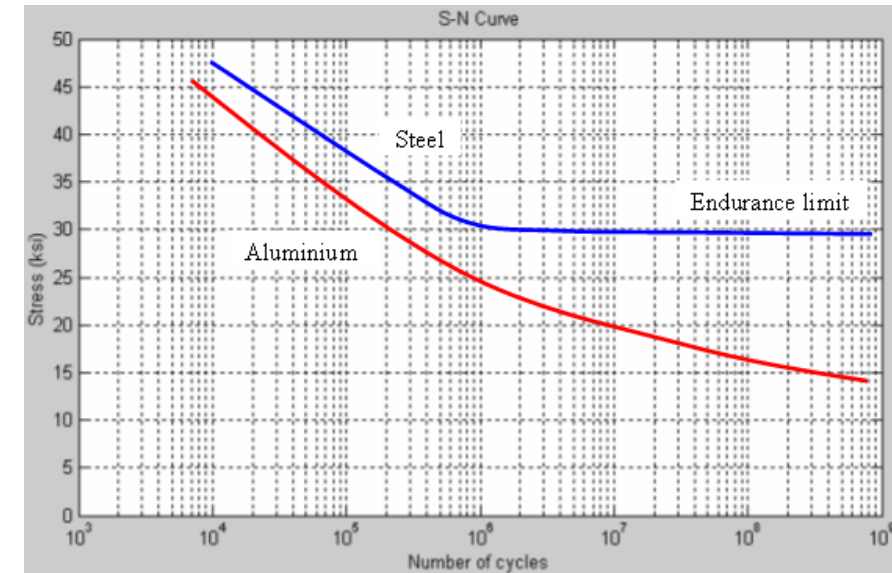
- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

- **Fatigue cycling occurs for repetitive loads often lower than yield or ultimate strength - cracks form and propagate through system until sudden fracture**
 - Examples:
 - Spacecraft tanks that undergo multiple pressurization cycles
 - Mechanisms that are repeated frequently (e.g., solar array gimbals, robotics motors)
 - Endurance Limit: Stress amplitude under which fatigue will not occur
 - Fatigue Life: Number of cycles of a load a structure can sustain without fatigue
 - Defined in S-N (Stress vs # of cycles) curves
 - FEA can support fatigue analysis, but typically needs to be validated with testing
- **Fatigue and durability testing:**
 - Purpose: Assesses long-term material performance under repeated or cyclic loads
 - Strategy: Applies repeated loading cycles to components to test for fatigue



S-N curve

Other Testing

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

• Mass properties testing:

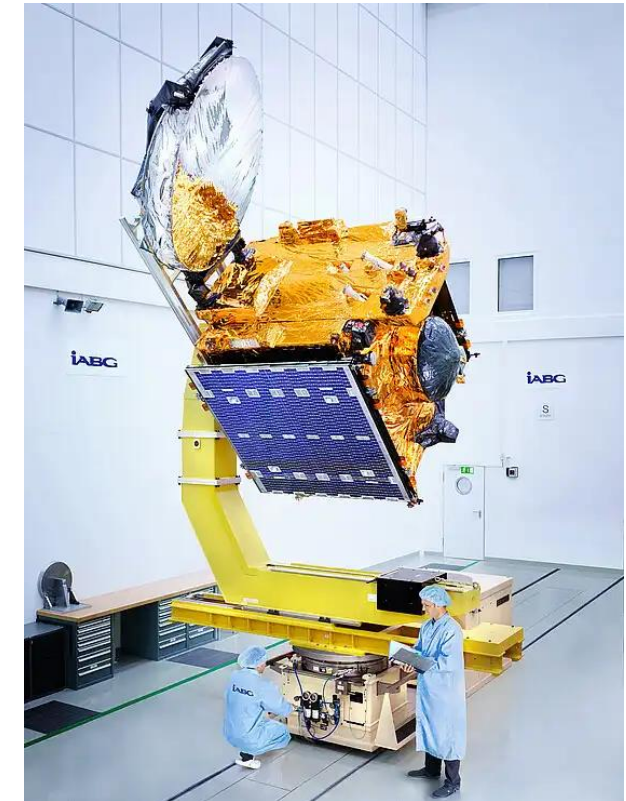
- Purpose: Ensure total mass, CG, MOI, POI are within specifications
- Strategies:
 - Mass: Uses precision scale or load cell
 - CG: Spacecraft is mounted on a fixture that allows it to pivot until it balances (minimal torque); location where it balances is used to determine the CG
 - MOI and POI: Spacecraft is placed on a spin table and rotated around each principal axis; angular response (frequency and acceleration) are measured and used to calculate MOI and POI for each axis

• Life testing:

- Purpose: Assesses durability of spacecraft components for expected operational environment and mission duration
- Strategy: Components are subjected to mission-like conditions (loads, vibrations, temperatures) over an extended period, typically ~1.5 to 2x life for margin
 - Can do accelerated testing at higher loads, but requires careful assessment to ensure this is representative

• Bakeout:

- Purpose: Reduces outgassing, prevents contamination of spacecraft to itself (contamination control) and other planetary bodies (planetary protection)
- Strategy: Spacecraft is elevated to high temperature (~100 to 125C) in vacuum (typically part of TVac testing); post bakeout inspection ensures cleanliness is met



Spacecraft mass properties testing

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and
Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

Mechanical Subsystem Design Process

Mechanical Subsystem Design Process

Overview

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

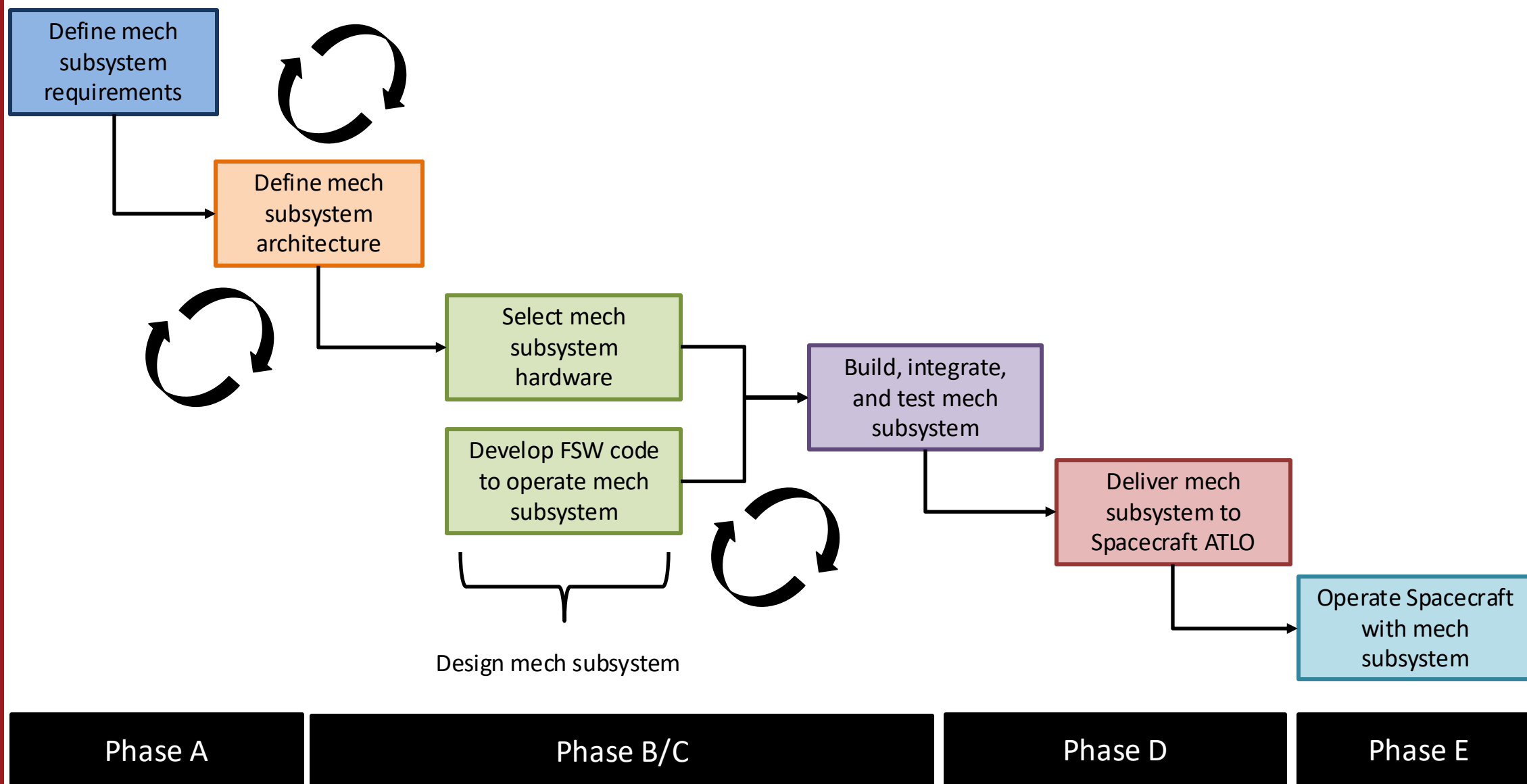
Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies



Mechanical Subsystem Design Process

Requirements

Mechanical

Introduction

• Overview

• Examples

Constraints & Reqs.

Design

• Structures

• Materials

• Mechanisms

• Configuration

• Loads and

Environments

• Mass Estimation

• Trade Studies

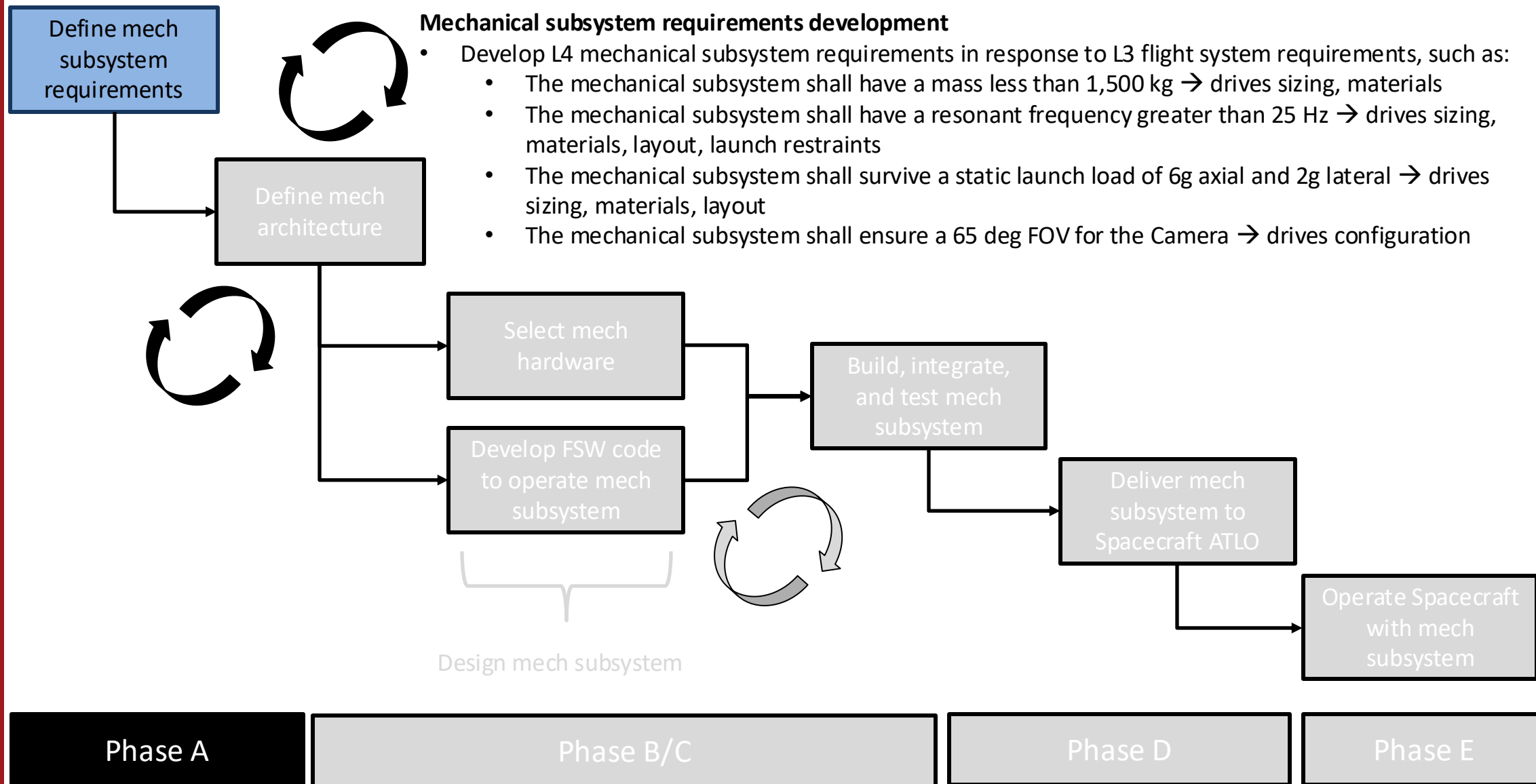
Analysis and Test

Design Process

Case Studies

Mechanical subsystem requirements development

- Develop L4 mechanical subsystem requirements in response to L3 flight system requirements, such as:
 - The mechanical subsystem shall have a mass less than 1,500 kg → drives sizing, materials
 - The mechanical subsystem shall have a resonant frequency greater than 25 Hz → drives sizing, materials, layout, launch restraints
 - The mechanical subsystem shall survive a static launch load of 6g axial and 2g lateral → drives sizing, materials, layout
 - The mechanical subsystem shall ensure a 65 deg FOV for the Camera → drives configuration



Mechanical Subsystem Design Process

Architecture

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

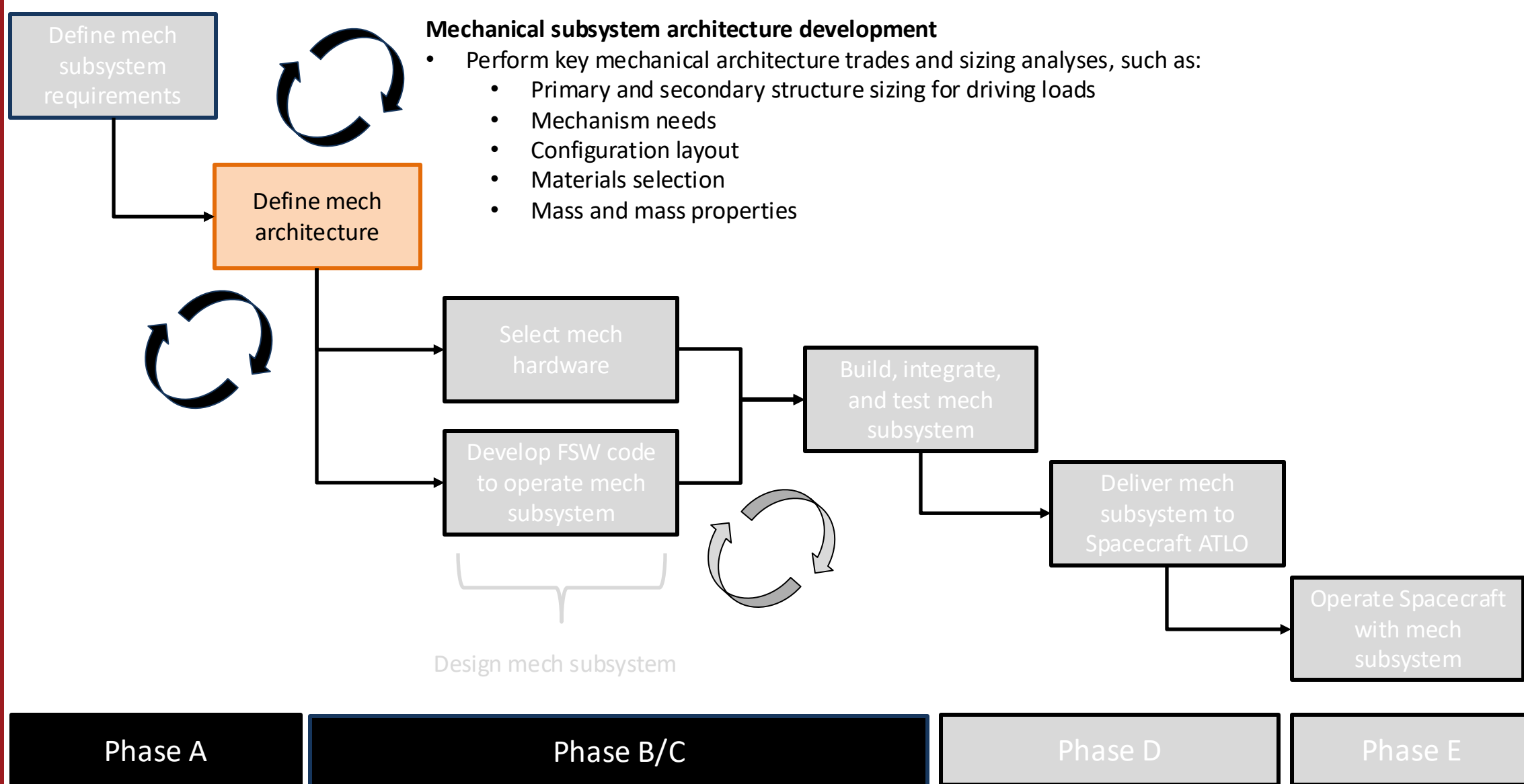
Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies



Mechanical Subsystem Design Process

Design

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

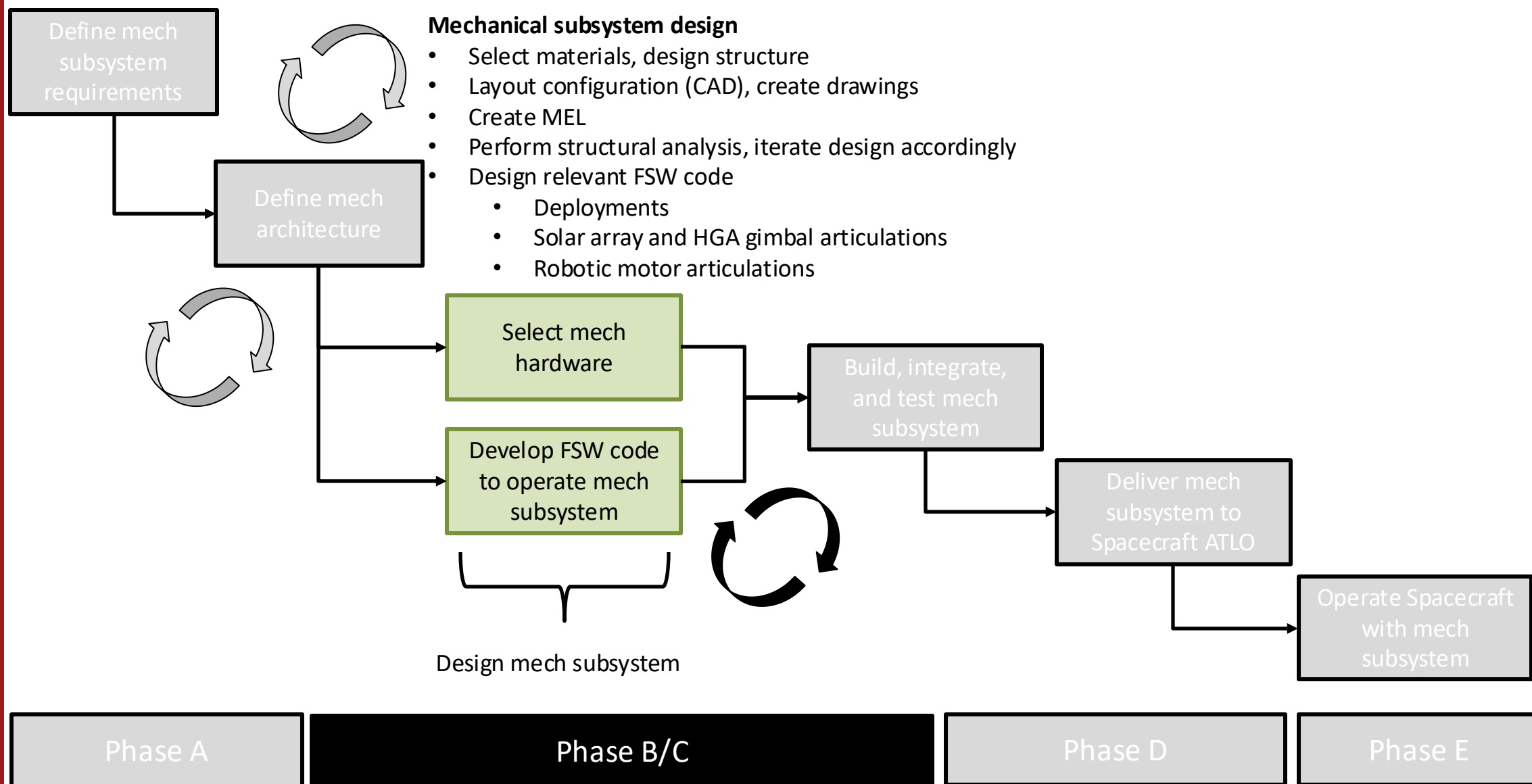
Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies



Mechanical Subsystem Design Process

Build and Test

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

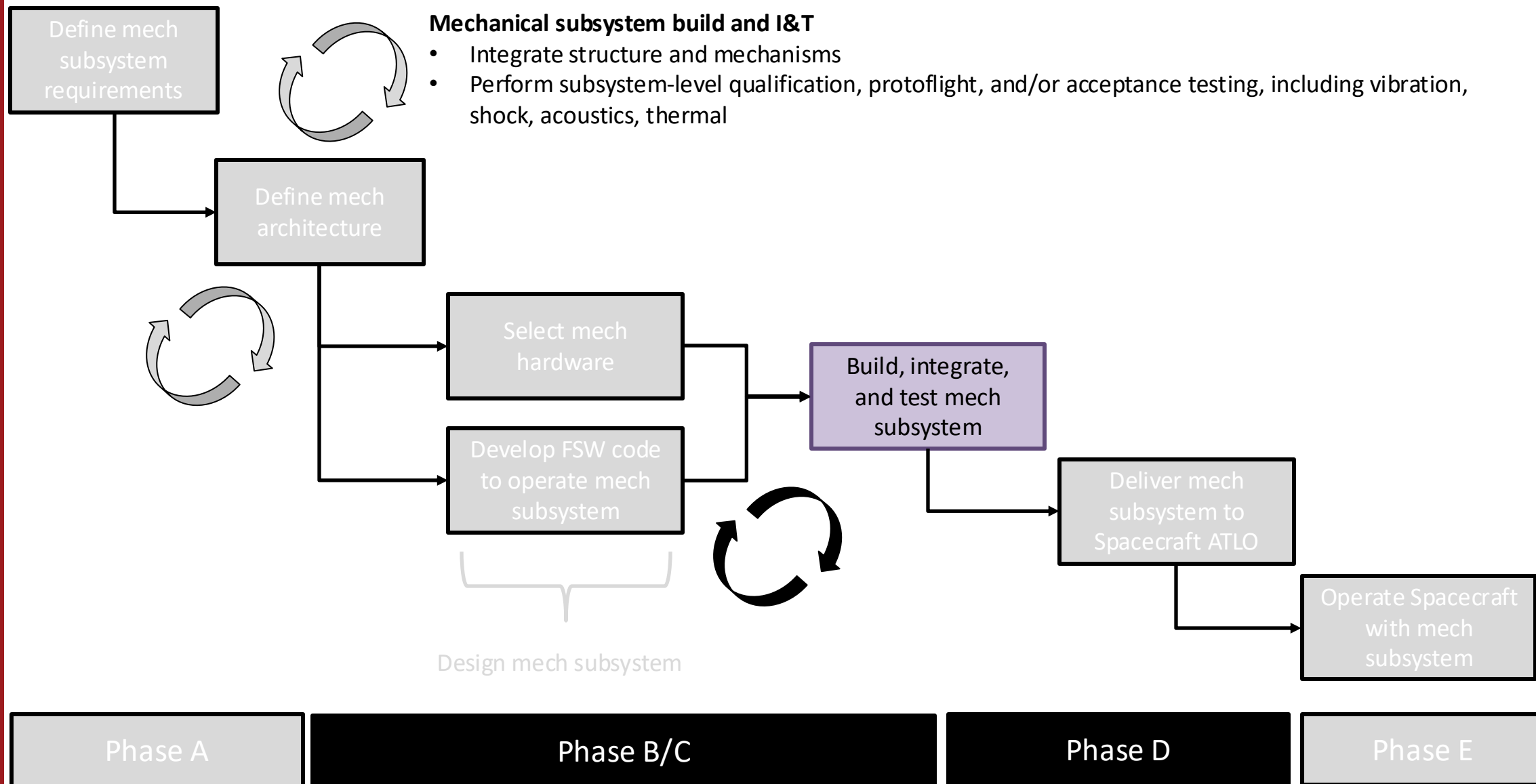
Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies



Mechanical Subsystem Design Process

Deliver

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

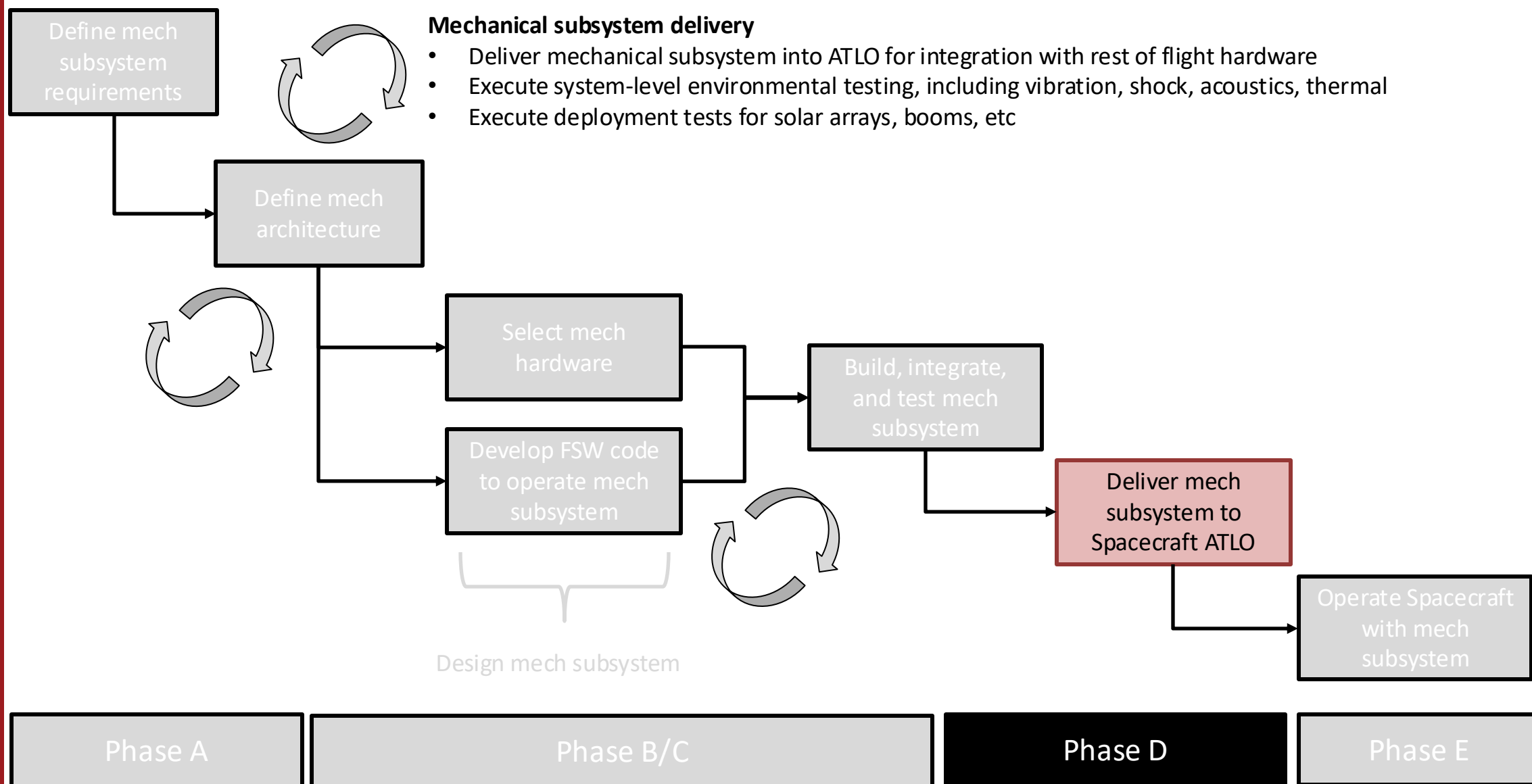
Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies



Mechanical Subsystem Design Process

Operate

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

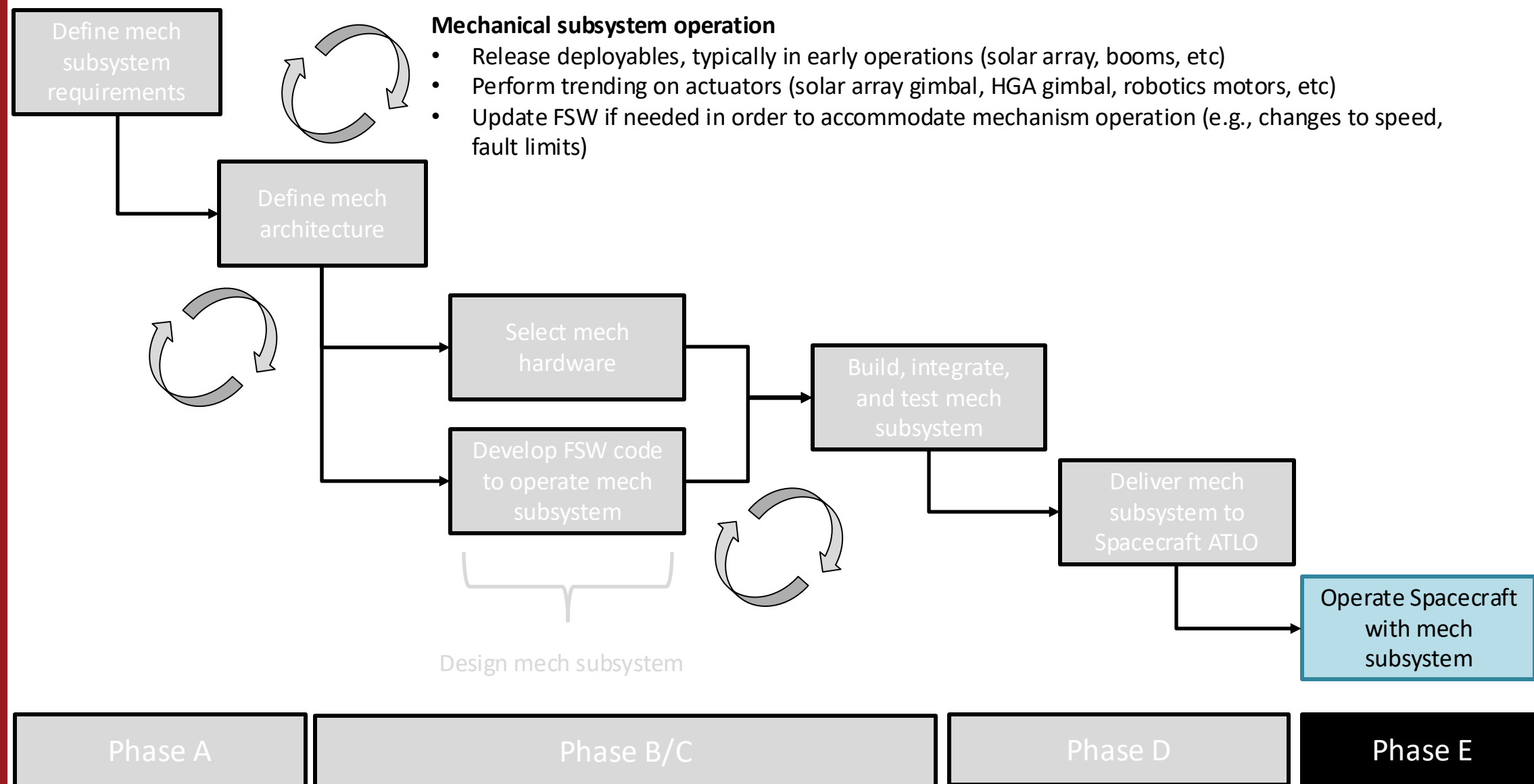
Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies



Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

- Structures
- Materials
- Mechanisms
- Configuration
- Loads and
Environments
- Mass Estimation
- Trade Studies

Analysis and Test

Design Process

Case Studies

Case Studies

Case Study: Beagle 2

Mechanical

Introduction

- Overview
- Examples

Constraints & Reqs.

Design

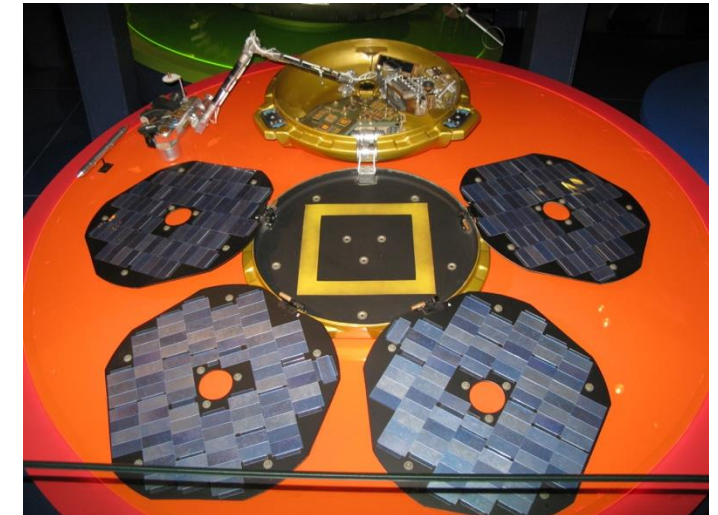
- Structures
- Materials
- Mechanisms
- Configuration
- Loads and Environments
- Mass Estimation
- Trade Studies

Analysis and Test

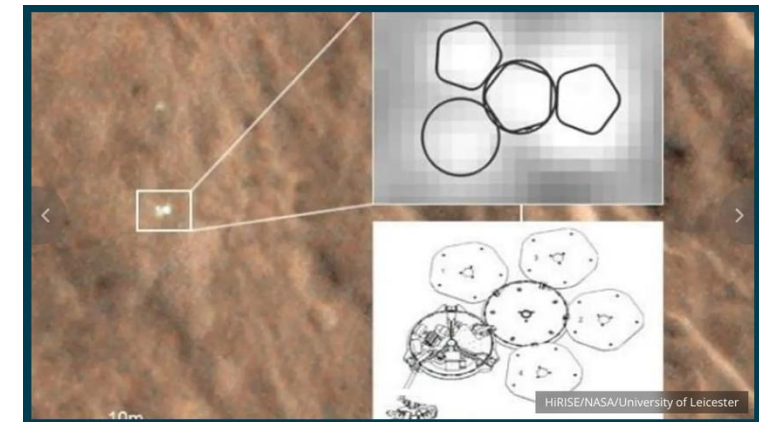
Design Process

Case Studies

- **Beagle 2 was an ESA Mars probe launched in 2003 as part of Mars Express**
 - Successfully separated from Mars Express, but failed to establish communication after landing
- **Key failure:** One or more solar panels did not fully deploy, likely obstructing the antenna
- **Causes:**
 - **No communication redundancy:** Relied on a single communication path, which became unusable due to the deployment issue
 - **Landing challenges:** The EDL sequence had unexpected atmospheric conditions, potentially damaging solar array deployment hardware
 - **Deployment complexity:** The success of communications depended on complete solar array deployment
 - **Limited testing:** Tight budget constraints limited extensive testing at realistic conditions, particularly of critical deployments after relevant environments
- **Lessons learned:**
 - **Redundant comm paths:** Multiple communication paths should be included to prevent a single point of failure
 - **Robust deployment systems:** Minimize dependencies in deployment sequences
 - **Enhanced testing for critical systems:** Allocate budget and schedule resources to enable comprehensive testing at realistic conditions, especially for high-risk operations like EDL and deployments
 - **Improved environmental modeling:** Ensure accurate and adaptive atmospheric models for Martian conditions



Intended deployment



Assumed actual deployment